

Does the formulation of a unified field theory require a fundamental change in the model of reality?

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Abstract

In this article, I present a model of reality that allows both relativistic and quantum effects to be described using a single model. The proposed model unifies these two descriptions by combining the concept of absolute motion in four-dimensional space with the effect of relative motion in the observed three-dimensional subspace. Describing the body directly as a wave—made possible by combining these motions—allows us to describe relativistic effects as well as electromagnetic interactions and inertial mass, which are treated as different aspects of the interactions of the waves described in this way. The new idea describes well-known phenomena, but it changes our understanding of them, leading to the prediction of new phenomena that could be the basis for experiments to determine which of the mechanisms responsible for relativistic effects—whether described by the theory of relativity or in this article—are correct. Several possibilities for experimentally verifying the proposed approach are presented, at least one of which can be implemented immediately. Moreover, this does not undermine the results and predictions of the theory of relativity or quantum mechanics but rather explains the mechanism behind phenomena that were previously considered the foundation of current models.

1 Introduction

The theory of relativity in its current form accurately describes physical phenomena, but it can be said that it is incomplete. Although numerous tests have confirmed the predictions of both special and general relativity, there are many areas where the theory of relativity encounters limitations or problems. We can mention, for example,

1. Singularities that cause physical quantities to take on infinite values under certain conditions. In such cases, it is impossible to describe the phenomena.
2. The inability to combine the theory of relativity with quantum mechanics and the so-called vacuum catastrophe causes enormous discrepancies between the theoretical predictions of the quantum mechanics model and astronomical observations.
3. The problem of the existence of dark matter and dark energy was introduced to explain the discrepancies between the theoretical calculations and the results of astronomical observations.
4. Problems with further developing the theory of relativity, such as the inability to unify the descriptions of electromagnetic and gravitational interactions, or the aforementioned difficulties in combining the theory of relativity with quantum mechanics.
5. Difficulties in understanding the relative motion of bodies resulting from the limitations on relative velocity or Mach's principle.
6. The inability to describe the Universe in a coordinate system independent of the observer, which may repeat the mistake made by Ptolemy, who, by using a description based on the observer's coordinate system (located on the Earth's surface), created an erroneous model of the Universe.
7. The lack of a mechanism to explain the similarity between the dimensions of time and space and the inconsistency with the well-established fact that phenomena take the simplest possible form; therefore, a four-dimensional reality should be based on four-dimensional Euclidean space.

The inability to resolve these and other problems within the existing model has led many authors to seek alternative solutions outside the previously used schemes, in which they discuss variants of the description of reality, most often based on four-dimensional Euclidean space [1-44]. However, these issues have not yet been satisfactorily resolved.

1.1 Goal of the work

In this paper, I discuss the possibility of resolving the above issues. Since existing methods have failed for more than 100 years, I would like to propose exploring an alternative approach to solving these problems.

When we develop a new theory to describe certain areas of physics, we naturally assume that known phenomena and verified observational results serve as the basis for the model. However, owing to specific aspects of the observational process—which we are not yet aware of at a given stage of the model's development—the observed effects sometimes differ from their actual form. One example is the retrograde motion of the planets observed in the firmament, which for 15 centuries was regarded as the result of actual retrograde motion of the planets in the Universe. This led to the introduction into the science of the concept of epicycles—which is now forgotten, but for the 15 centuries, it considered a brilliant explanation for the strange motions of the planets. Moreover, the observed retrograde motion of the planets was not the result of actual retrograde motion of the planets but merely the result of the fact that the reference frame from which the observations were made was not stationary—as was assumed at the time—but moved around the Sun just like all the other planets.

Despite being based on a flawed model of the Universe, the epicyclic model produced results consistent with observations, albeit at the cost of applying complex mathematics based on artificially assumed mechanisms—in this case, epicycles. The complicated model of the universe was therefore not the result of the complex nature of the phenomena but rather the result of adopting an erroneous model as the basis for constructing the model.

The necessity of using complex mathematics in the theory of relativity or quantum mechanics and the acceptance of certain phenomena as postulates is very reminiscent of the philosophy behind the construction of the epicyclic model—despite the completely different nature of the phenomena being described. This suggests that accepting the observed shape of reality in these models as its actual image may be a repetition of Ptolemy's error, who also made an assumption of a similar nature.

In this article, we attempt to construct from scratch a parallel model of reality that describes known phenomena. However, in the process of creating the model, we place great emphasis on the mechanism of observation and the relationship between the observed form of reality and its form as predicted by the new model. By combining these two factors—the model as a representation of actual reality and observation, that is, the relationship between its actual and observed forms—we attempt to explain the existing problems associated with the model of reality.

1.2 Method

The currently accepted models were developed on an ongoing basis, mainly by adding new elements to an existing framework, using new experimental findings and assumptions—sometimes adopted on an ad hoc basis—as the foundation for the model to explain the experimental results as simply and quickly as possible, for example, in the interpretation of the results of the Michelson–Morley experiment [67]. As a result, we have ended up with tools that

accurately describe phenomena but are highly complex and often downright incomprehensible and can be described only through the use of mathematics. I would dare to say that current models serve more as a manual for quantitatively describing phenomena and predicting certain additional effects than as an explanation of the actual mechanism behind these phenomena. Perhaps this is simply the nature of reality, and we must accept it. However, it would be appropriate to first check whether it is possible to create a simpler and more general, parallel description of reality—especially since research and work on models of reality to date have provided us with a great deal of new information that was unknown at the beginning of the 20th century when the foundations of current models were created, and which, if rearranged and supplemented with a few concepts that tie the entire model together, might lead to the creation of a unified and comprehensible description of all known phenomena.

Since efforts over the past century have not yielded satisfactory explanations, we will try here to examine the known issues from a much broader perspective.

First and foremost, we must take into account the fact that the reality we observe may differ from its actual form because of the process of observation, in which—as already mentioned—we currently assume, following Ptolemy's example, that phenomena appear exactly as we record them in our frame of reference. Such an assumption has already led to an erroneous model once before, so one might ask the following question: are we not currently witnessing a repetition of the same error in today's models?

To reduce the probability of making an error by potentially using results distorted by the observation process as the basis for the model, we base the model's assumptions not on specific phenomena but on certain general properties verified over the years, which, moreover, are not treated in this work as postulates but as hypotheses that should be explained in the new model by appropriate, as yet unknown, mechanisms.

The proposed properties on which we base our reasoning are as follows:

- Reality is four-dimensional.
- The speed of light is constant for all inertial observers.
- When two inertial bodies observe each other, time dilation is observed in accordance with the condition of conservation of the spacetime interval.
- Massive bodies have wave properties.
- The four properties listed above are supplemented by a rule that applies throughout all stages of model development, which states that every phenomenon takes the simplest possible form—although the process of observation may distort our perception of that phenomenon. This means that at every stage, we choose the simplest solution, even if it seems to contradict our intuitive understanding of the phenomenon.

In turn, all other observed, known, and well-described properties—which have thus far been taken as the basis for building the model—will now serve merely as milestones that we should reach in subsequent stages of the model's development.

Thus, concepts such as motion, time, space, fields, energy, momentum, and charges—and, consequently, the relationships based on these properties, such as the equations of Lagrange, Hamilton, and Maxwell, or the laws of conservation of energy and momentum—are not initially applied here. They will emerge or become derivable in subsequent stages of the model's development once these concepts are defined within the new model.

1.3 What to Expect from the New Approach

In this work, we would like to find common mechanisms underlying theories that have hitherto been treated as distinct. Understanding these mechanisms will allow us to introduce initial conditions and constraints for a mathematical description that leads to the derivation of known relationships, which are, however, treated—in accordance with the new model—as different aspects of the same, common mechanisms. However, introducing new tools for quantitatively calculating the effects of the phenomena described by existing models is not the purpose of this work. The primary goal is to introduce mechanisms that will help us understand the origins of the models used to date. A better understanding of the fundamentals of reality should make it possible to explain the limitations of existing models, as well as new limitations that have not yet been identified, and to supplement our description of reality with new tools capable of describing new phenomena that cannot be described by today's models. In essence, this is not about modifying or overturning existing theories. Rather, it is about creating a new branch of physics that explains the origins and mechanisms of the phenomena underlying the phenomena on which current models are based but which those models cannot explain.

This new approach may give rise to many misunderstandings because of its entirely new treatment of reality compared with the existing spacetime model. All the concepts related to the spacetime description can be compared with the current state of knowledge only after a transition from the description of the phenomena in the new model to the spacetime description. At the stage of the model's construction itself, any attempts to apply equations and concepts adapted to the spacetime description will lead to contradictions and misunderstandings.

To better convey the purpose of the work to be done, I use a simple comparison. If we have maps of sea areas, we use two coordinates to determine the locations and distances of continents and islands, which in practice are used to plan routes, calculate travel times and costs, etc. However, the question of the origin of islands—why there are individual islands or archipelagos—and why the distances between them are what they are—cannot be explained using a two-dimensional model such as a map. Adding a third dimension—depth—does not alter the existing data on the map, but it does help us understand what islands and continents are. We obtain a three-dimensional image of underwater mountain ranges; when these intersect with the water surface, they reveal the outlines of islands and their relative distances, measured along the water surface, as depicted on maps. The third dimension may be included on maps, but only as supplementary information; the primary and most useful result for us is the intersection of the three-dimensional structure of the seabed with the two-dimensional surface of the water. However, the introduction of a third dimension does not change the map of the oceans. It is crucial only for understanding the origin of islands and the shapes of archipelagos and continents or for predicting the emergence of new islands or the disappearance of existing ones as a result of tectonic movements.

In the case of the simplest four-dimensional reality, namely, Euclidean space (not four-dimensional spacetime, in which dimensions change as a function of motion), we obtain a four-dimensional description of phenomena analogous to the aforementioned three-dimensional underwater mountain ranges, and by intersecting this four-dimensional space with a three-dimensional structure representing our space, we obtain an image of three-dimensional space and time as well as the relative motion of bodies within this three-dimensional space (similar to the creation of a two-dimensional map of the sea resulting from the intersection of the aforementioned underwater mountain ranges with a plane representing the surface of the

oceans). Only this three-dimensional structure, with time as we know it, can serve as the basis for describing observed reality using the concepts of time, space, charges, masses, fields, etc., that we are familiar with.

This is merely a rough example intended to provide a general overview of the basic idea, which will prove to be more complex and abstract in detail; however, the current aim is to convey the main idea of the work in a relatively clear manner and to avoid misunderstandings arising from the use, in the evaluation of the work, concepts specific to a spacetime description, which is, however, inappropriate from the perspective of the model presented in this work.

2 A new concept of time and special dimensions

When Einstein announced his theory in 1905—according to which the existence of the ether was not necessary to explain relativistic problems—he seemingly put an end to the ether problem.

In the late 1920s, a complete set of information became available that allowed scientists to return to the ether hypothesis in a form different from the ether as it had been conceived at the end of the 19th century. However, the recent successes of the theory of relativity and the new, promising, and imagination-stirring concepts of quantum mechanics at that time meant that this possibility went unnoticed. However, the emergence of the problems mentioned above and the fact that they have remained unsolved for many decades should prompt us to also consider this overlooked possibility.

The new set of information I mentioned consists essentially of just two properties of reality:

1. The reality is four-dimensional.
2. Massive bodies have wave properties.

The essence of the concept presented here is the assumption that reality is built as a four-dimensional space in which bodies move along certain trajectories at a constant speed—for simplicity, we will assume this speed to be equal to one [47,48, 58, 64,66 and all others between 47 and 66]. Particles observe each other through interactions, and we interpret the directions of propagation of these interactions in E4 as spatial dimensions.

And that is essentially it. Everything else consists of conclusions that specify the specific properties of particles, motion, and the mutual observation of bodies.

The new four-dimensional space is neither spacetime nor the kind of space we are familiar with. It is described by certain distances that do not possess predefined properties of time or space. Thus, we describe the paths of particles in this space using the term “trajectory” rather than “world line,” as is the case in spacetime.

First, I present a preliminary outline of how the model works (this will be described in more detail later).

According to the approach proposed here, bodies move through four-dimensional space along their trajectories and observe each other.

We assume that the bodies move along a trajectory at a speed of unity, so the distance traveled by a body is, from its perspective, a measure of time for that body, and the trajectory can be interpreted as the time axis of its coordinate system. The lack of any restrictions on the directions of the trajectories of bodies means that, for each body, a different direction in four-dimensional

space can be interpreted as the time axis of the body's coordinate system—or, in other words, as the time dimension of the observer's spacetime coordinate system.

In turn, bodies moving along parallel trajectories observe each other as being at rest. Bodies moving along trajectories inclined relative to the trajectories of other bodies observe a change in their distance from those bodies with time—that is, the motion of the bodies—whereby such motion is not the result of any additional motion relative to space but solely of the angle of inclination of the bodies' trajectory relative to the observer's trajectory. In four-dimensional space, none of the directions are distinguished. Thus, the angle between the trajectories of the bodies is relative and can be determined only in relation to the trajectories of other bodies. Therefore, the observed motion of the bodies resulting from this inclination of their trajectories is also relative.

Furthermore, the constant velocity of bodies in four-dimensional space allows a particle to be described directly as a wave.

This is, of course, only a sketch of the approach presented below. However, to write about time, space, interactions, etc., we must first define the relationship between the objective dimensions of the new four-dimensional space and the observed spacetime dimensions of the coordinate systems of bodies.

We begin by introducing a condition that allows us to distinguish the spatial coordinates of the observer's reference frame.

We know that bodies do not perceive space as such. Bodies observe the motion of surrounding bodies, and on the basis of these observations, observers construct a picture of space. In the new four-dimensional space, we interpret certain directions as spatial directions. However, what are these directions?

We currently assume that space dimensions are simultaneously directions in space-time, along which interactions propagate.

And now I would like to rewrite that last sentence in two ways:

1. The observer observes the motion of bodies along spatial directions.
2. The observer observes the motion of bodies along the directions of the propagation of interactions.

Both statements seem to be equivalent because, in the current model of reality, these directions are identified.

However, can we treat both statements as equivalent?

Sentence 1 links the directions of propagation of interactions with the model of reality. This means that spatial directions determine the directions of propagation of interactions.

On the other hand, sentence 2 links the direction of propagation of interactions with the particle sending the interactions, which allows us to become independent of the current model of reality. According to sentence 2, spatial dimensions do not determine the directions of the propagation of interactions; rather, by observing the motions of bodies in four-dimensional reality along the directions of the propagation of interactions, we interpret these directions as spatial dimensions of the observer's coordinate system. Interpreting spatial dimensions not as directions belonging to reality but as directions dependent on bodies will allow for greater freedom in defining the space that constitutes reality, for example, as Euclidean space, provided that the model is

supplemented with an observation process such that, regardless of the assumed shape of reality, the observed image of reality is Minkowski's space-time.

This new interpretation of spatial dimensions as a consequence of sentence 2 is considered here as the method—or perhaps only one of the potential methods—for solving all the problems mentioned above.

2.1 Initial assumptions of the model

Assuming that spatial dimensions are interpreted as directions in reality determined by the directions of interactions sent by particles, we can essentially start by accepting any space as a model of reality.

Let us start with the 19th-century model of space but increase the number of dimensions by 1.

So:

1. Let us initially assume that reality is a four-dimensional Euclidean space E_4 . The dimensions describing E_4 define certain distances, but they do not have a preassigned meaning of time or spatial distance.
2. In this space, time T is defined, which is not the fifth dimension but a parameter that allows all events to be ordered, and time T is identical for all bodies. [47, 58, 59, 60, 62-64, 66]
3. In space E_4 , there are bodies that move at an absolute speed $V = 1$ —initially, we assume that the bodies move only in rectilinear paths. In this case, the distance traveled by the body—let us denote it as s —is equal to $s = VT$, and since $V = 1$, $s = T$. Thus, the distance traveled by the body in space E_4 is also a measure of time measured by the particle clock, and in the case of rectilinear motion, the particle time is equal to time T : $s = t = T$. Consequently, the particle can interpret its trajectory in E_4 as the time axis of its coordinate system and the distances measured in E_4 along its trajectory as time distances.
4. Each body sends interactions into a three-dimensional subspace orthogonal to the trajectory, i.e., the time axis of the particle's coordinate system. Since the directions of propagation of interactions are interpreted by the observer as spatial dimensions of the observer's coordinate system, the three-dimensional subspace E_4 orthogonal to the trajectory of the observed particle determines three directions interpreted by the observer as spatial dimensions of his reference frame for observing a specific particle. Thus, according to the new model, the directions of propagation of interactions determine the spatial dimensions of the observer, and the distances measured by the observer along the directions of propagation of interactions are interpreted by the observer as spatial distances. The idea of spatial dimensions becomes the property of observation and ceases to be the property of reality.
5. At this stage, we will only assume that signals are transmitted in the observer's system at a speed equal to unity, $V=1$.

The mechanism ensuring the constancy of the speed of light is presented in the appendix: [Mutual observation of bodies](#)

Assumptions 1--5 define the initial ideas for building the model. These ideas will be developed and modified later in the article.

On the basis of the above conditions, we can draw several conclusions at this stage:

2.2 Simple conclusions

2.2.1 Time dilation and speed limit the speed of light

The problem of mutual observation of bodies in accordance with the terms and conditions set out at points 1--5 is shown in Fig. 1.

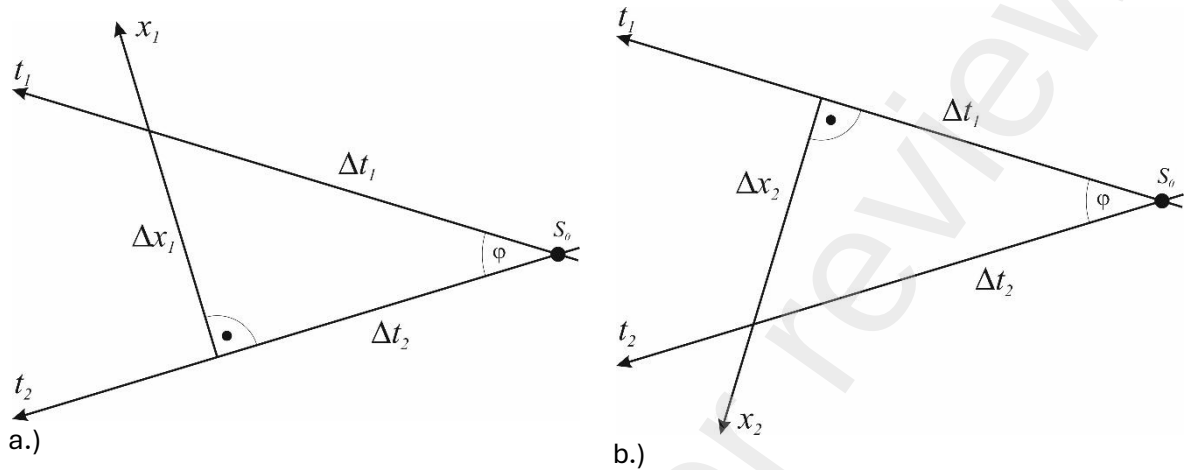


Fig. 1 Mutual observation of two bodies in E4. The directions in E4 interpreted as space-time coordinates of the observer depend on the trajectory of the observer (interpreted as its time dimension) and the trajectory of the observed body, where the directions perpendicular to this trajectory are interpreted as space directions, i.e., described by the space dimensions of the observer's coordinate system. In fig. a.) observer x_1, t_1 observes body t_2 , in fig. b.) observer x_2, t_2 observes body t_1 .

The observers observe the velocity of the observed body V as the sine of the angle between their trajectories (the time axes of their coordinate systems).

$$V = \frac{\Delta x_i}{\Delta t_i} = \sin \varphi \leq 1 \text{ where for Fig. 1, } i = 1, 2 \quad (1)$$

and they also observe the relativistic time dilation in the system of observed bodies, according to equations (2,3):

For observer t_1 observing body t_2 , we have the following:

$$\Delta t_2 = \Delta t_1 \cos \varphi = \Delta t_1 \sqrt{1 - V^2} \quad (2)$$

For observer t_2 observing body t_1 , we have the following:

$$\Delta t_1 = \Delta t_2 \cos \varphi = \Delta t_2 \sqrt{1 - V^2} \quad (3)$$

The described time dilation effect is the result of observation and has nothing to do with the actual time dilation. For the mutually observed time dilation to become actual time dilation, the observer whose system time is to actually slow down must change the direction or value of their velocity. A detailed description of the process can be found in the appendix [Mechanism to change the time](#).

In turn, the speed of light is described by a separate mechanism, which ensures that regardless of the speed of the body relative to the observer, the observed speed of light is always equal to unity. A diagram showing the trajectory (not the world line) of light in the case of the two observers

is shown in Fig. 2, while the exact principle of signal (quantum) propagation is described in the appendix [Mutual observation of bodies](#).

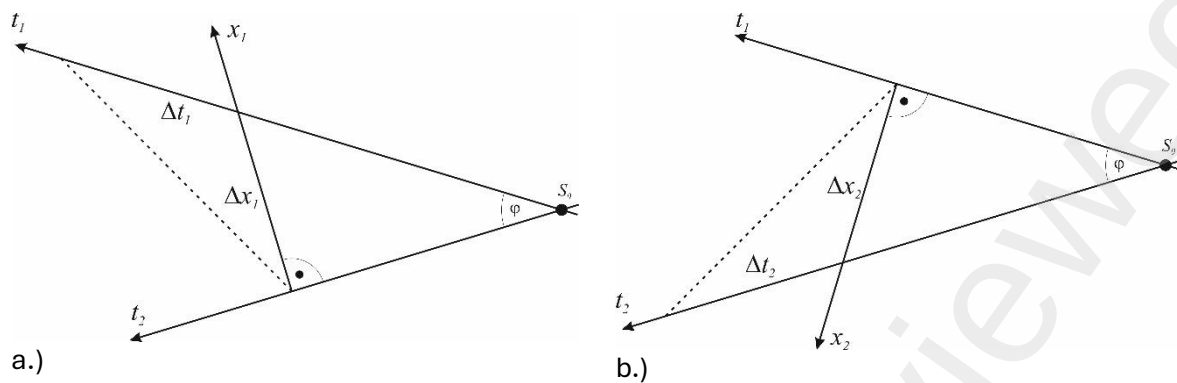


Fig. 2. Trajectory (not world line) of a signal sent by an observed body to an observer. The trajectory of the signal is shown as a dotted line. In Fig. a), observer x_1, t_1 receives a signal sent by observer t_2 , and in Fig. b.) vice versa – observer t_2, x_2 receives a signal sent by observer t_1 . The spatial axis of the observer is inclined at a right angle to the time axis of the observed body. Since $\Delta x_i = \Delta t_i$ the observed signal velocity is always equal to $V_l = \frac{\Delta x_i}{\Delta t_i} = 1$ regardless of the angle φ determining the relative velocity of the bodies according to formula (1).

Conclusions regarding the speed of bodies:

1. The relative velocity of bodies is determined by the angle between the trajectories of bodies in E4.
2. The concept of bodies at rest refers to bodies moving in E4 at an absolute velocity $V = 1$ along parallel trajectories.
3. The limitation of the particle velocity to a value of 1 resulting from formula (1) does not constitute a limitation on trajectories but rather only on observations. All trajectories are permitted, but bodies moving along trajectories inclined to each other at an angle $\geq \frac{\pi}{2}$ cannot observe each other through the exchange of interactions.
4. The motion of the light in E4 shown in Fig. 2 causes the observed speed of the light to always be equal to unity, regardless of the angle φ , which is a measure of the relative speed of the body. It is easy to see that the trajectory of a hypothetical rocket traveling at the speed of light $V = 1$, i.e., at an angle $\varphi = \frac{\pi}{2}$ relative to the observer's trajectory in E4, will be different from any trajectory of light that can be recorded by the observer, also observed as equal to unity $V_l = 1$. Thus, the motion of light and the motion of massive bodies are described by different mechanisms.

As seen from the example presented, the relative motion of bodies and the motion of light quanta are different phenomena, but with the help of our measuring devices, we record them as if they were exactly the same phenomenon—similar to the observed motion of the sun and moon, which are different motions observed as if they were identical. As mentioned earlier, a detailed explanation of the mechanism ensuring the constancy of the speed of light is described in the appendix [Mutual observation of bodies](#)

2.2.2 Observation of multiple bodies, or why we observe reality as Minkowski spacetime

A comparison of observations of bodies in E4 space and Minkowski spacetime is shown in Fig. 3

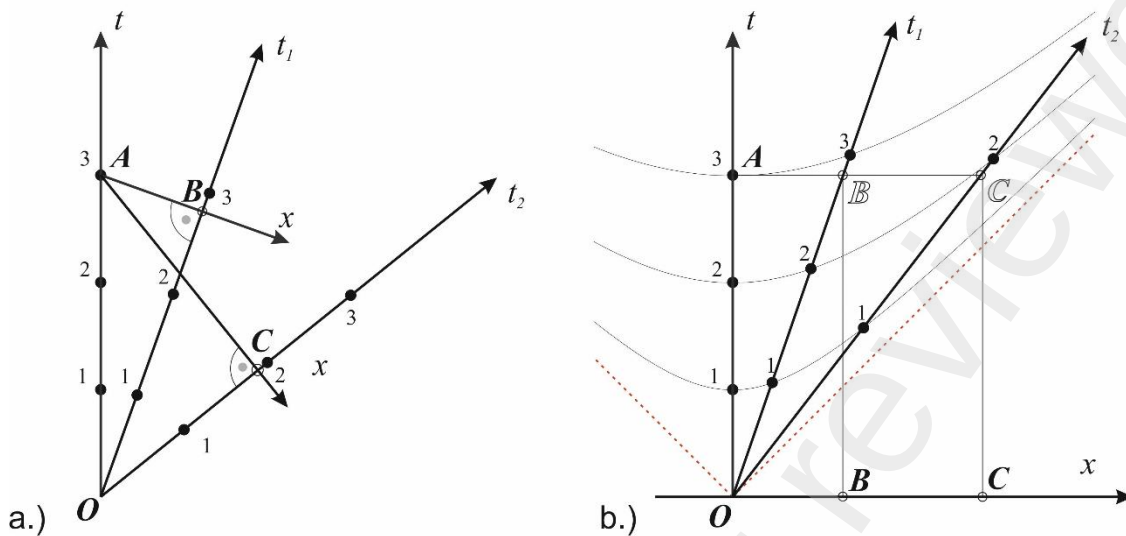


Fig. 3. Comparison of observations of bodies in E4 space and in Minkowski spacetime. Observer t observes two bodies t_1 i t_2 . In space E4 – fig. a.) – the observer observes each of the bodies along a direction perpendicular to the trajectory (time axis of the coordinate system) of the observed body, which is the direction in which signals (radiation quanta) are emitted by the observed body. Since the observer registers the motions of bodies along the directions of the propagation of interactions, the observer interprets the directions of the propagation of interactions as his spatial dimensions. By observing different bodies in E4, the observer interprets different directions in E4 as their spatial dimensions and measures the distance from body t_1 along segment AB and the distance from body t_2 along segment AC . All axes, both time and space, are on the same scale, and relativistic relationships (1--3) result directly from the geometric relationships between the axes of the coordinate systems. Fig. b.) shows a description of the same problem in Minkowski spacetime. An observer who is convinced that spatial distances are always perpendicular to the time axis of his coordinate system plots the measured distances AB and AC on the same space axis of his coordinate system. In this case, plotting the time axes of the observed systems requires stretching the time axes of these systems. A comparison of figures a.) and b.) shows that segments OB and OC (of the time axes) in Fig. b.) are stretched in relation to their actual length, as shown in Fig. a.). This causes the axes of the coordinate systems of the observed bodies to be deformed, and the values of the coordinates of the systems in motion cannot be read from the figure—here, it is necessary to use additional mathematical dependencies resulting from the principle of conservation of the space–time interval.

As shown in the figure, the description of observations in E4 space preserves the scale of the dimensions of all bodies (Fig. 3a), which allows for the description of the problem of mutual observation of bodies from any coordinate system, even one not related to any observer. However, we achieve this at the cost of accepting the seemingly absurd assumption that when we observe different bodies, we interpret different directions in E4 as our space dimensions and are unaware of the difference between these directions. However, equally absurd is the assumption that the dimensions of bodies are deformed as a function of velocity, which describes the same relativistic effects in Minkowski spacetime, but here, no one asks questions because for more than 100 years, we have become accustomed to this assumption.

Moreover, the observer cannot determine that the distances from both bodies are measured along different directions in E4 because there are no objective reference points in E4 against which the observer can obtain information that the directions he interprets as space dimensions differ for the observation of different bodies.

Thus, the observer is convinced that the distances from bodies t_1 and t_2 are measured along the same spatial direction, and since it is assumed that in four-dimensional space-time, the spatial dimensions are perpendicular to the observer's time axis, he plots the measured segments AB and AC from Fig. 3a on the same spatial axis of their reference frame, perpendicular to its time axis. With this type of presentation, the time axes of the t_1 and t_2 systems must be stretched, as shown in Fig. 3b. This is the image of space-time that we imagine when our observations are being conducted.

As a result, when we observe the motion of bodies in four-dimensional Euclidean space, we are convinced that we live in Minkowski's four-dimensional space-time.

3 Combination of Relativity Theory and Quantum Mechanics

3.1 Definition of a body as a wave

Assuming constant particle motion in E4 allows us to define a particle directly as a wave. We can therefore treat space as an elastic medium. To derive the wave equation, we need to know the properties of the medium. In the case of space E4, we do not yet know these properties—they must be determined first. The only information we have at this point is Planck's constant and the value of the electric charge. Until the properties of space as an elastic medium are determined, these values will have to suffice.

What we can currently use is the wave function equation describing a particle, which is the solution to Schrodinger's equation:

$$\Psi = e^{-\frac{i}{\hbar}(Et - px)} \quad (4)$$

To express this equation in E4 space, we must rewrite it in E4 coordinates, whereas it is currently written in spacetime coordinates—here, x and t . After conversion to the E4 coordinate system:

$$\begin{aligned} \Psi = e^{-\frac{i}{\hbar}(Et - px)} &= e^{-\frac{i}{\hbar}(m_0 \frac{dt}{ds} t - m_0 \frac{dx}{ds} x)} = e^{-i \frac{m_0}{\hbar} (\frac{d(t^2)}{2ds} - \frac{d(x^2)}{2ds})} = e^{-i \frac{m_0}{\hbar} (\frac{d(t^2 - x^2)}{2ds})} = e^{-i \frac{m_0 d(s^2)}{\hbar 2ds}} = \\ &= e^{-i \frac{m_0 2ds}{\hbar 2ds}} = e^{-i \frac{m_0}{\hbar} s} \end{aligned} \quad (5)$$

the function takes the form:

$$\Psi = e^{-i \frac{m_0}{\hbar} s} \quad (6)$$

where:

$$\omega_0 = \frac{m_0}{\hbar} \quad (7)$$

What can we write down as:

$$\Psi = \cos(\omega_0 s) - i \sin(\omega_0 s) \quad (8)$$

where s is the length of the path traveled by the particle along its trajectory in E4.

If our wave is to describe a real particle, we should equip it with the following additional features:

1. We apply amplitude $A(r)$ to the particle, which determines the amplitude of the particle in the direction $\vec{r}$ perpendicular to the particle's trajectory in E4 (the exact shape of the amplitude is not yet known). While in equation (4) the spatial coordinate of any of the bodies— x —was perpendicular to the time axis of the distinguished observer, the direction $\vec{r}$ in E4 is defined by any direction in the subspace perpendicular to the trajectory of the body described by equation (8). Since the trajectories of different bodies may differ, the directions $\vec{r}$ will also be directions in different subspaces perpendicular to the trajectories of the various bodies. Thus, the equation does not describe the wave using the spacetime coordinates we are familiar with, but rather using the Euclidean coordinates of E4 space:

$$\Psi = A(r)[\cos(\omega_0 s) - i \sin(\omega_0 s)] \quad (9)$$

2. The particle equation should include the motion of the particle in E4 along the direction s :

$$\Psi = A(r)[\cos(\omega_0 s) - i \sin(\omega_0 s)] - is \quad (10)$$

3. The value of the function should describe both the densification and the rarefaction of space, which should result in the function depending on the sign of the amplitude. This can be achieved, for example, in the following way:

$$\Psi = \pm A(r)\{1 - [\cos(\omega_0 s) - i \sin(\omega_0 s)]\} - is \quad (11)$$

4. Since the propagation direction of the interactions in the new model is perpendicular to the trajectory of the body, the interactions should propagate along the ridge of the wave. To enable the propagation of interactions along the ridge of the wave, let us add the phase velocity along the ridge of the wave, which is also equal to unity:

$$\Psi = \pm A(r)\{1 - [\cos(\omega_0 s \mp \omega_0 r) - i \sin(\omega_0 s \mp \omega_0 r)]\} - is \quad (12)$$

As seen from point 4 of the wave phase sign, the particles can not only send interactions but also receive them. Moreover, in chapter 2, point 4, I initially wrote that particles send interactions in a direction perpendicular to their trajectory in E4. In practice, information transfer consists of a particle sending a signal along its ridge, and then, at the point of contact between the ridges of the particles, the signal passes to the ridge of the particle receiving the signal and then propagates along its ridge but is now perpendicular to the time axis of the body receiving the signal. Only the combination of sending and receiving signals leads to the conclusion that the observer interprets (under certain conditions) the directions perpendicular to the direction of the trajectory of the body observed in E4 as the space axes of his coordinate system. A detailed description of the mechanism is shown in the appendix [Mutual observation of bodies](#).

3.2 New definition of a particle's proper time

The description of a particle as a wave in the form of formulas (6) and (8) or in its extended version (12) means that we can define the particle's proper time $\Delta t'$ with an accuracy of the particle's wavelength as the number of periods of vibration n of the particle on the section of the trajectory Δs traveled by the body in space E4.

In the case of inertial motion, when a body moves along a rectilinear trajectory at a constant speed relative to E4, the number of wave vibrations representing the measure of proper time per unit length of the trajectory traveled by the wave is constant; thus, the proper time of the wave $\Delta t'$ can

be measured as the length of the path in E4, i.e., $\Delta s = \sqrt{\sum_{i=1}^4 \Delta a_i^2}$, or as the number of wave periods n .

$$\Delta t' = \Delta s = \sqrt{\sum_{i=1}^4 \Delta a_i^2} \cong n \frac{2\pi}{\omega} \quad (13)$$

where a_i – Euclidean coordinates describing space E4. Hence, initially, the interpretation of the particle's proper time as a measure of the distance traveled by the particle along its trajectory was adopted (Fig. 1).

However, identifying the path traveled by a body in E4 with the time axis of its coordinate system is possible only for inertial motions. For noninertial motions, the relationship between the number of vibrations and the length of the trajectory depends on the shape of the trajectory. For example, on the E4 path connecting two points A and B, for one trajectory, the particle time is n cycles, and for another trajectory, it is m cycles, with the highest number of cycles being achieved on a rectilinear trajectory. While a nonzero distance between points on the world line in Minkowski spacetime cannot correspond to zero elapsed time, in the case of a wave trajectory in E4, it is theoretically possible to consider such segments of the trajectory where the phase of the wave does not change; i.e., the number of wave cycles, and thus, the measure of proper time on this segment, can be equal to zero.

More on this topic can be found in the appendix [Mechanism to change the time](#).

3.3 Unified theory of fields, definition of electric charges and mass

The possibility resulting from the new model of reality to describe a particle directly as a wave defining two types of space deformation—compression and rarefaction—and the interpretation of oscillations in three-dimensional space as rotations of deformation in four-dimensional space (8,12) provide the basis for understanding the phenomenon of both electric charges and the source of inertial mass and probably also gravitational mass.

Thus, the mutual interactions of waves, which are rarefactions and compressions of space, justify electrical interactions, whereas the rotation of space deformations in four-dimensional space, combined with the definition of velocity change in E4 as a change in the direction of motion of a rotating deformation in E4, justifies the existence of inertial forces.

3.3.1 Electric charges

The wave function describes the deformation of space. The sign of the amplitude determines whether the deformation is positive or negative, which we can interpret as oscillating compression or rarefaction of space.

The influence of the amplitude sign on the shape of the wave describing the particle is shown for the actual value of the function in Fig. 4 (arbitrary units).

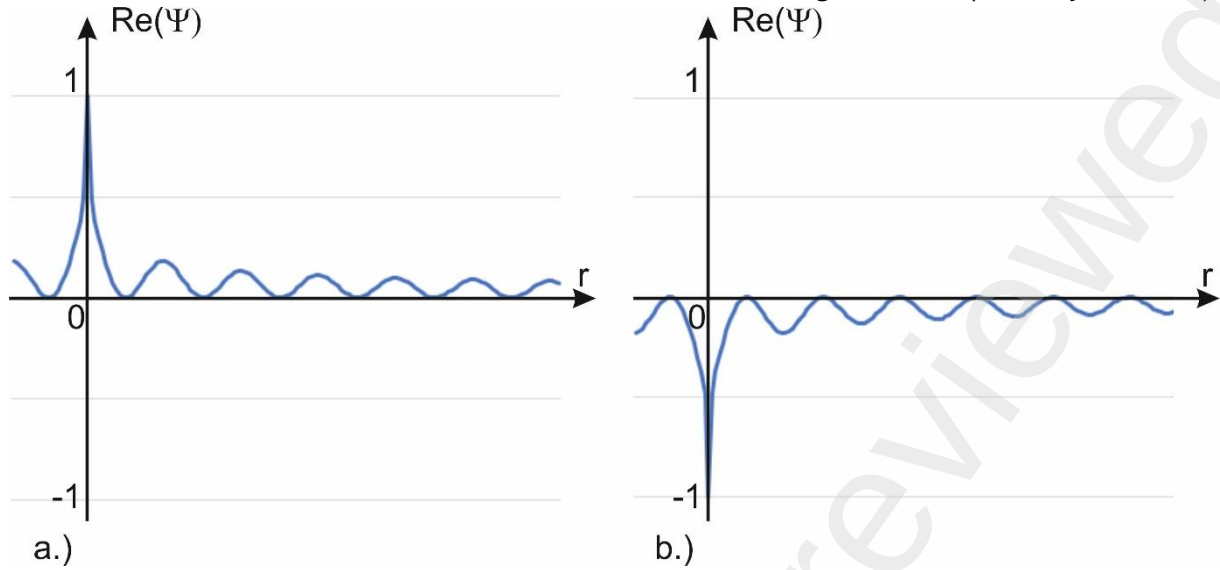


Fig. 4 Example distribution of the real part of the wave amplitude describing a particle (12) for a particle with positive amplitude (Fig. 4a) and negative amplitude (Fig. 4b).

The deformation of a medium is always associated with a certain amount of energy. At present, we do not know the exact relationship between the deformation of space and energy; thus, for now, we can only write the following:

$$E = f(\Psi) \quad (14)$$

where $f(\Psi)$ is a certain, as yet unknown, function determining the energy accompanying the deformation of space.

By describing a particle as a wave, we assume that space is an elastic medium. For very small disturbances in an elastic medium, the energy of such a disturbance can be described by the following formula:

$$E = k \frac{x^2}{2} \quad (15)$$

where x denotes the deformation of the medium. We do not know the value of the coefficient k for space, and we do not know how the energy of space deformation behaves for larger deformations. However, on the basis of the laws we know, which often work similarly on both macro- and micro-scales, we can attempt to describe the mechanism of interactions between particles described as waves, for now only for small space disturbances, via relation (15). In practice, this approach is most likely applicable only at very large distances from the center of the particle, but to demonstrate the mechanism of the phenomenon, we apply it to describe the entire wave.

Since the function Ψ denotes the amplitude of the function, i.e., simply the deformation of space, the energy of space deformation E_4 for the complex on segment dr should be¹:

¹ For the purposes of demonstrating the mechanism, formulas (16-24) are given only for the two-dimensional case; finding the coefficient k requires a full analysis of the multidimensional problem.

$$dE(r, t) = k\psi(r, t)\psi^*(r, t)dr \quad (16)$$

where k is the coefficient linking space deformation with energy. At present, the coefficient k is still unknown; it will probably be possible to obtain it from the analysis of electrical and gravitational interactions.

In that case, the energy value of a single particle is as follows:

$$E(t) = \int_0^\infty k\psi(r, t)\psi^*(r, t)dr \quad (17)$$

Now, let us assume that for a fixed moment in time, for example, $t = t_0$, the energy of the particle is equal to one:

$$E(t_0) = \int_0^\infty k\psi(r, t_0)\psi^*(r, t_0)dr=1 \quad (18)$$

If we now consider two particles with identical amplitude distributions $A(r)$ and identical frequencies ω_0 , then if these particles are infinitely distant from each other, the space disturbances from these particles do not add up; thus, the total energy of two particles infinitely distant from each other is equal to the sum of their energies, i.e., the energy of these two particles is equal to 2. This energy does not depend on the sign of the particle's amplitude; i.e., its value does not depend on whether we consider two particles with identical or opposite signs of amplitude $A(r)$.

$$E(t_0) = \int_0^\infty k\psi(r, t_0)\psi^*(r, t_0)dr + \int_0^\infty K\psi(r, t_0)\psi^*(r, t_0)dr=2 \quad (19)$$

If we now place two particles with identical parameters and identical amplitude distributions $A(r)$ at the same point $r = 0$, then for particles with the same signs, the amplitudes of both particles add up, and we have the following:

$$E(t_0) = \int_0^\infty k2\psi(r, t_0)2\psi^*(r, t_0)dr = 4 \int_0^\infty K\psi(r, t_0)\psi^*(r, t_0)dr = 4 \quad (20)$$

If we spread the particles to infinity, their energy decreases from $E(t_0)=4$ at point $r = 0$ to $E(t_0)=2$ at infinity, which means that the force defined as the derivative of energy with respect to distance is equal to:

$$F = \frac{dE}{dr} \quad (21)$$

and has a positive sign, and the particles repel each other.

If we now place the same particles as above at point $r = 0$ but with opposite signs, the amplitudes will subtract and result in zero deformation, and then:

$$E(t_0) = \int_0^\infty k[\psi(r, t_0) - \psi(r, t_0)][\psi^*(r, t_0) - \psi^*(r, t_0)]dr = 0 \quad (22)$$

In this case, spreading the particles to infinity increases the total energy of the particle system and changes the energy of the particle system from $E(t_0)=0$ at point $r = 0$ to $E(t_0)=2$ at infinity; i.e., the force defined as the derivative of energy with respect to distance is equal to:

$$F = -\frac{dE}{dr} \quad (23)$$

and has a negative sign, and the particles attract each other.

On the basis of the above considerations, we can determine the energy of a system of two arbitrary particles with the same or opposite signs and separated by a distance R as follows:

$$E(R) = k \int_{r=0}^{r=\infty} \int_{t=0}^{t=T} (\psi_1(r, t) + \psi_2(r + R, t)) (\psi_1(r, t) + \psi_2(r + R, t))^* dr dt \quad (24)$$

Formula (24) was derived for the assumption of very small space disturbances, which most likely do not correspond to the actual picture of the phenomenon, but even in such a “provisional” form, it allows us to illustrate the mechanism responsible for particle interactions, which manifests itself to us in the form of electric fields.

In the above considerations, we assume that a particle is a combination of vibrations in a single plane. In reality, the term “particle” described by equation (24) should be understood as the single, smallest possible disturbance in space. In practice, real particles are composed of many such small elements. More complex vibrations are also likely possible. However, for the presentation of the model, I adopt the simplest possible solution.

3.3.2 Particle mass

Model E4 describes a particle as a combination of longitudinal and transverse wave oscillations (12), which in practice results in the rotation of space deformation in the plane of motion. We can see this in Figure 5, where the wave is a vortex moving in the direction determined by the plane of rotation, here determined by the directions s and y . If a compression or rarefaction rotates, it must have angular momentum. In the figure, angular momentum is denoted by the letter L . Note that in the case shown in Fig. 5, particle interactions are possible only in the direction determined by the r axis. If the dimensions s, y and r denote the dimensions of four-dimensional space E4, then a particle defined in this way exchanges interactions only in the plane formed by the direction r and the fourth dimension x , which is not shown in this figure. Therefore, if interactions occur equally in all three dimensions, we can, for example, compose a real particle from three elementary particles vibrating in three mutually perpendicular planes. Additionally, since the vibration phase of a single particle has two directions—toward and away from the center of the wave—to ensure interaction in all three dimensions as well as signal transmission and reception, a “real” particle would have to consist of at least six such elements. Perhaps this problem could also be solved by introducing more complex multidimensional vibrations or an additional fifth dimension, but I will leave this topic for possible discussion at a later date.

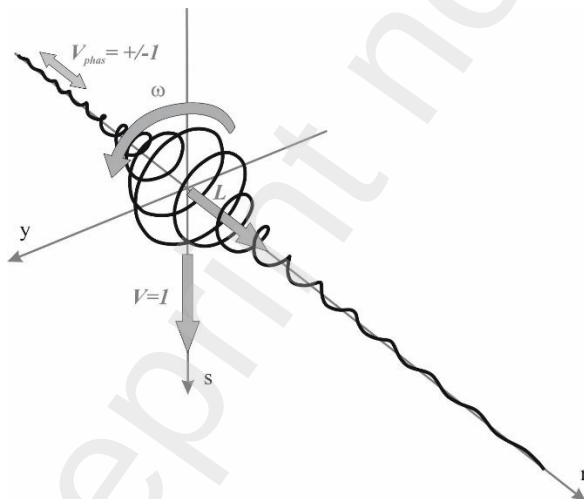


Fig. 5 Shape of the wave described by equation (12). The wave moves along direction s at speed $V = 1$ relative to the space treated as a wave carrier. It is a combination of transverse and longitudinal vibrations, which causes the deformation of space to rotate at an angular velocity ω in the plane determined by the direction of wave motion s and the direction perpendicular to this direction y . A rotating deformation has angular momentum. Any change in the direction of wave

vibrations corresponding to a change in the velocity (rotation in the s, r plane) or velocity direction (rotation in the y, r plane) causes a change in the direction of angular momentum, which counteracts the change in direction or relative wave velocity. Interestingly, the momentum of the wave remains constant only when the direction of the wave motion changes in the s, y plane, but because a “real” particle should probably consist of six such elementary waves vibrating in different planes, this effect has no assigned meaning at this time.

Now, changing the value or direction of the body's velocity means rotation of the body's wave in the s, r or y, r plane in E4, respectively (Fig. 5), which entails a change in the direction of the angular momentum vector $\vec{L}$. Thus, the body resists the change in direction because the angular momentum of the particle interferes with it. From mechanics, we know that the resistance a particle will offer before changing direction will increase the faster the particle rotates; i.e., the inertia of the particle will be proportional to the rotation frequency ω (i.e., to the frequency of oscillation of the particle wave). The mass of a particle is described by the following formula:

$$m = E = \hbar\omega \quad (\text{here, } c=1) \quad (25)$$

The gravitational mass is equal to the inertial mass, so the proportionality coefficient linking the inertial mass to the frequency of disturbance rotation in space E4 is Planck's constant.

If we recall the formula from mechanics to describe rotational motion energy:

$$E = \frac{I\omega^2}{2} = \frac{(I\omega)\omega}{2} = \frac{L\omega}{2} \quad (26)$$

where I denotes the moment of inertia of the body.

A comparison of equation (26) with equation (25) reveals that Planck's constant makes sense as the smallest possible momentum of a particle.

The source of the inertial mass of a particle is therefore the principle of conservation of angular momentum of a particle in E4. The value of mass is therefore the unit angular momentum multiplied by the frequency of rotation of the disturbance in E4, which opposes the change in the trajectory of the particle in E4, i.e., the change in the value or direction of velocity.

In terms of gravitational mass and gravitational interactions, the source of gravitational interactions is most likely related to the tendency of a system of particles to reduce the total rotational motion energy of space deformation, for example, by reducing the total angular momentum of the system by increasing the angles between the angular momentum vectors of individual particles. This is not fully explained, but an analysis of the energy of the particle system on the basis of the philosophy applied in equation (24) should clarify the issue.

So:

According to the model presented here, electric charge is the result of space deformation consisting of compression or rarefaction of space, whereas the mass of bodies is the result of the rotation of space deformation in the plane determined by the direction of motion of deformation in E4 and one of the directions perpendicular to this direction.

4 Does the new model solve unsolved scientific problems?

The aim of introducing this new description of reality, in addition to simplifying the description, is to attempt to resolve previously unsolved problems. The introduction listed certain problems that existing models have been unable to solve. In the following subsections of this chapter, I demonstrate the extent to which the problems mentioned in the introduction have been resolved in the new model.

However, solving existing problems sometimes leads to the emergence of new ones, which are far fewer in number than in the current model; nevertheless, resolving them will require the development of additional concepts and solutions that we have not encountered in previous models—as discussed in more detail in the next chapter.

4.1 Singularities

In the model that describes reality using Minkowski spacetime, singularities occur at the speed of light and at the boundaries of black holes. In the Minkowski model, these singularities are a property of reality.

Let us start by looking at a singularity related to the speed of light.

The diagrams of the mutual observation of two bodies in E4 space are shown in Figure 1. The requirement that the direction interpreted by the observer as a spatial direction must be perpendicular to the trajectory of the observed body in E4 (the time axis of the observed body's coordinate system) follows the principle of conservation of the space-time interval for such an observation. The mechanism that necessitates the use of such a condition is described in Appendix 7.1 [Mutual observation of bodies](#).

The description of the mutual observation of the bodies shown in Fig. 1 makes sense only if the time axes of both bodies and the observer's spatial axis form the sides of a right triangle, provided that the angle φ between the trajectories t_1 and t_2 of the observing bodies is less than a right angle: $\varphi < \frac{\pi}{2}$. This is because the mechanism for transferring information—that is, observation—as described in detail in the appendix 7.1 [Mutual observation of bodies](#) allows information to be transferred only between bodies moving along trajectories inclined to one another at an angle less than a right angle $\varphi < \frac{\pi}{2}$. This is because the transmission of information here consists of the disturbance first traveling along the ridge of the wave, and then, at the point where the ridges of the waves from the signal-sending body and the signal-receiving body intersect, the signal transfers to the ridge of the receiving wave and then travels along its ridge. In the case of the trajectories of bodies inclined to each other at a higher angle—here, $\varphi \geq \frac{\pi}{2}$ —the point of intersection of the wave ridges moves away from the center of both waves at a speed equal to or greater than the signal propagation speed (phase velocity) along the ridge of the wave that emits the signal. Thus, the transmitted signal will never catch up to the ridge of the receiving wave; i.e., it will not be transferred to the ridge of that second wave. Consequently, although Fig. 1 can also be presented for the case $\varphi \geq \frac{\pi}{2}$, this is not a physical case because it no longer describes the observation of bodies.

Although in E4, all trajectories are permitted—including those that are inclined relative to the observer's trajectory by angles equal to or greater than $\frac{\pi}{2}$ —for angles greater than or equal to $\frac{\pi}{2}$, it is not possible to determine the observer's actual spacetime coordinate system for the

observation of that particular body. This limitation refers to the inability to define a spacetime coordinate system; in other words, the inability to observe bodies moving along trajectories for which the angle of inclination relative to the observer's trajectory is greater than or equal to a right angle— $\varphi \geq \frac{\pi}{2}$. However, this does not imply a physical limitation that prevents such a trajectory from being achieved. Moreover, in the theory of relativity, the impossibility of observing a body reaching the speed of light is equated with a physical limitation of the phenomenon resulting from the properties of reality.

By allowing for motions that cannot be observed or described in the observer's coordinate system, the E4 model clearly expands the model's capabilities compared with those of previous models.

When the body's velocity equals the speed of light—which, in E4, corresponds to the body following a trajectory perpendicular to the observer's trajectory—a singularity appears in the spacetime description.

To explain the origin of the singularity in the E4 model more clearly, let us consider the example of accelerating a rocket of constant mass—the problem is described in more detail in [65]. The problem is illustrated in Fig. 6.

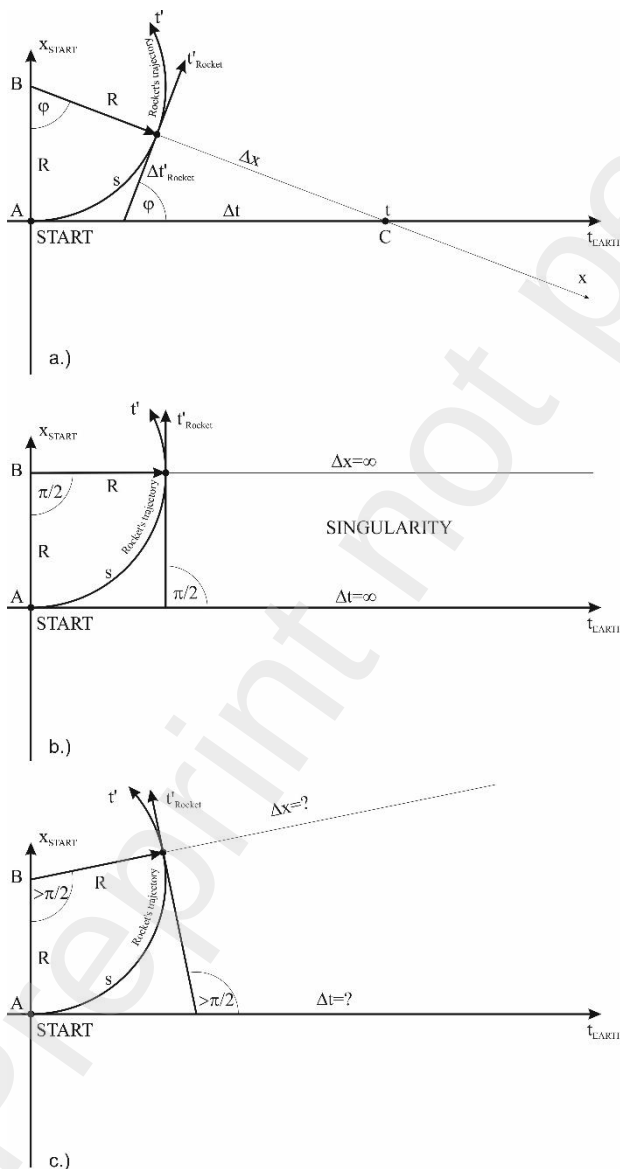


Fig. 6 A rocket of constant mass moves with constant acceleration, which in E4 is equal to the inverse of the radius of curvature of the rocket's trajectory $a = \frac{1}{R}$.

Figure a.) shows the trajectory of a rocket moving at a speed of $V = \frac{\Delta x}{\Delta t} = \sin \varphi$ as observed by an observer on Earth. As the rocket's speed increases, time and distance also increase, tending toward infinity. Therefore, an observer will watch the rocket accelerate, but will never observe that the rocket has reached a speed of $V = 1$.

Figure b.) shows the moment when the rocket reaches a speed of $V = 1$. From the passengers' perspective, nothing will change—they will feel the same acceleration and perceive the passage of time in the same way. However, from the perspective of an observer on Earth, a singularity occurs in such a case. This singularity is not a limitation on the rocket's motion, but rather a limitation on the observation process. The rocket can continue to accelerate as before—as shown in figure c.)—and the passengers will not feel any additional effects. Simply put, from now on, none of the signals sent by the rocket will be able to reach an observer on Earth. The motion of the rocket along the trajectory shown in figures b.) and c.) cannot, therefore, be described using the idea of velocity or using spacetime coordinates.

In E4, an accelerated body moves along a curved path whose radius of curvature is equal to the inverse of the acceleration [65].

The coordinates of the observer's reference frame, determined in accordance with the principles shown in Fig. 1 for this case, are shown in Fig. 6a. The rocket's velocity is the sine of the angle between the tangent to the rocket's trajectory - t' and the observer's time axis. The speed of the rocket as measured by the observer is here equal to $V = \frac{\Delta x}{\Delta t} = \sin\varphi \leq 1$ and, by definition, is less than one, which in the E4 model is also the speed of light.

When the rocket reaches a trajectory inclined to the observer's trajectory at an angle $\varphi = \frac{\pi}{2}$ - fig. 6 b The values of the observer's spacetime coordinates determined for the observation of the rocket become infinite, and the case under consideration in the observer's spacetime coordinates becomes nonphysical. There is a singularity here. Since, in the observer's frame of reference, the distance to the rocket increases continuously over time and the observed velocity of the rocket approaches $V = 1$ asymptotically, the observer will continue to see the rocket's image fading to infinity. What cannot be described in the observer's spacetime coordinates can be described in the coordinates of E4 space, as shown in Fig. 6c. We can see that there are no obstacles here to the rocket continuing to change its trajectory. However, the observer is now unable to observe the rocket because none of the signals emitted by the rocket will reach the observer on Earth—the observer will now see only the rocket receding, as shown in Fig. 6a, and nothing more. From the passengers' point of view, nothing changes. Mass does not increase to infinite values, the time does not stop or start flowing in the opposite direction—they simply observe their surroundings just as we observe ours while driving down a winding road—some stars will disappear, and others will appear. However, going beyond the right angle will not cause the rocket to reach its destination faster. On the contrary—it's more likely that it won't reach anywhere at all, because by the time it gets to its destination, the planet we were heading for will be long gone. Here, we see a clear difference between the trajectory we use in E4 and the world line in Minkowski spacetime. For two bodies to collide, it is not enough for their trajectories to have a common point in E4—as is the case with world lines. The bodies must also meet there at the same absolute time—a time common to all bodies. I will discuss these issues briefly in Section 5.4.

With respect to the subject of speed, according to the new model, the motion of massive bodies and the motion of light quanta are distinct phenomena—described in the appendix 7.1 [Mutual observation of bodies](#). Therefore, it is not reasonable to compare these motions using the concept of speed. For example, the trajectory of a hypothetical rocket traveling at the speed of light in E4 is inclined relative to the observer's trajectory by an angle of $\frac{\pi}{2}$ - Fig. 6b, whereas the trajectory of a light signal between two measurement points in the observer's laboratory setup is inclined relative to the observer's trajectory by an angle of $\frac{\pi}{4}$. In this case, the observed speeds of the rocket and the light beam are identical, but their trajectories in E4 are different. Moreover, in the spacetime coordinate system of Minkowski spacetime, both velocities are identical, and their worldlines coincide (although there is a singularity on such a worldline for the rocket).

In E4, the trajectory of a hypothetical rocket traveling at the speed of light and the trajectory of light quanta are different trajectories corresponding to different types of motion and reaching different regions in E4. Moreover, according to Minkowski's spacetime model, both a hypothetical rocket traveling at the speed of light and a quantum of light emitted from the same point should, in theory, arrive at the same location at the same time. More on this topic is given in Section 5.4.1.

4.2 The combination of quantum mechanics with the theory of relativity and “vacuum catastrophe”

The direct description of a particle as a wave—Equation (12) and Fig. 5—in four-dimensional space, which can be represented in the new model as an elastic medium, changes our understanding of the concept of a field. In empty space, it was necessary to introduce a mechanism that would allow us to describe the values of the field. Owing to the absence of a medium—in this case, ether—a kind of ether substitute was employed, namely, the concept of the quantum vacuum, which in turn led, among other things, to the “vacuum catastrophe” paradox.

In the new model, we return to the concept of an elastic medium filling space, and we can describe a body directly as a wave that disturbs this medium (12). Thus, we can now describe a field as a disturbance of the medium generated by waves, which no longer requires the use of such substitutes. The deformation of space described by the wave equation is associated with deformation energy, and the result of the superposition of deformations originating from different waves is an interaction aimed at reducing the total deformation energy of the system of bodies/waves (24). In the case of single waves or the interaction of several waves, the shape of the wave influences the distribution of spatial deformations generated by the wave system. In such cases, local energy maxima and minima of the wave system arise both in the context of linear motions and in the case of rotational motions—such as those of an electron in an atom—which leads to the formation of local energy minima of the system that depend on the frequency of the waves, which in turn should be observed as quantum effects. In the case of bodies made up of a large number of waves, the spatial deformations caused by individual waves are averaged out, suggesting that constant forces are acting on the bodies; however, these forces are actually the result of the superposition of oscillations originating from all the waves that make up the interacting bodies. Therefore, there is no distinction here between bodies and fields because fields are, so to speak, an extension of bodies. To describe the effect of the fields, we can use relationships similar to equations (24) owing to the possibility of introducing an elastic medium, we no longer need to introduce the concept of an empty space filled with quanta possessing strange properties because the oscillations—which must occur throughout the entire space—are not the result of the existence of hypothetical quanta but are caused by the vibrations of particle waves. Since there is no need to fill the vacuum with particles capable of describing fields, the vacuum energy is limited solely to the sum of the vibrational energy of the medium derived from all the particles in space; in practice, this corresponds to the energy of the fields originating from all the particles. In this way, we avoid the “vacuum catastrophe” characteristic of the quantum vacuum

Moreover, the separation between quantum mechanics and the theory of relativity disappears because all quantum and relativistic properties are part of the same description that defines the interactions of particles described as waves. Field effects and quantum effects arise from the properties of the wave representing a particle (12), whereas relativistic effects arise from the mechanism by which disturbances are transferred between such waves—more on this topic in Chapter 4.4.

4.3 Dark energy and dark matter.

The problem of the existence of dark matter has not yet been explained within the framework of the presented model. However, the problem of dark energy is explained here as a consequence of describing bodies directly as waves moving in E4 at a speed of $V = 1$ and of the way in which

these bodies observe one another. It is possible that the as-yet-unexplained problem of dark matter could also be explained in a similar way.

The principle that states that when observing bodies, we treat the directions perpendicular to the trajectory of the observed body as the spatial dimensions of the observer's coordinate system justifies time dilation, and the same principle justifies Hubble's law [52, 58].

Fig. 7 shows an observer observing two bodies (body 1 and body 2), with the trajectories of the observer and observed bodies having a common origin, where all these bodies at time $T = 0$ met.

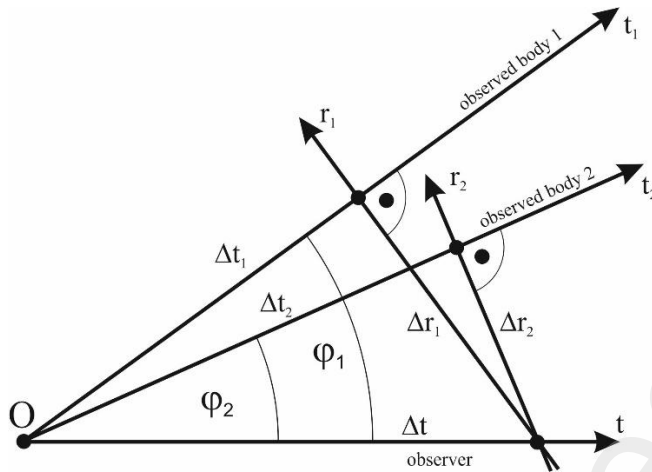


Fig. 7 An observer and two bodies moving along trajectories that have a common origin. When observing a body moving along a trajectory t_i ($i = 1, 2$), the observer interprets the direction r_i as the space dimension of his reference system. The lengths of the trajectories traveled by the observer and the observed bodies from point O to the current position of the bodies are measures of the elapsed times in the systems of these bodies. When the scheme is applied to describe the case when the point O is the Big Bang, the time Δt is the age of the Universe measured in the observer's system.

The observation of each body is represented by two right triangles: for the observation of the first body formed by the sides $\Delta t_1, \Delta r_1, \Delta t$ and for the observation of the second body by the sides $\Delta t_2, \Delta r_2, \Delta t$.

From these triangles, we obtain the following relationships:

$$\Delta t_i = \Delta t \cos \varphi_i = \Delta t \sqrt{1 - \sin^2 \varphi_i} = \Delta t \sqrt{1 - V_i^2} \quad \text{where } i = 1, 2 \quad (27)$$

describing the time dilation observed in body systems 1 and 2,

and equation:

$$V_i = \frac{\Delta r_i}{\Delta t} \quad (28)$$

Therefore, for bodies moving along rectilinear trajectories with a common origin, the observed velocity of a body is proportional to the observed distance from that body.

When applied to describe the motion of galaxies along trajectories with a common origin, i.e., Big Bang, this formula directly describes Hubble's law [52, 58]:

$$V_i = H \Delta r_i \quad (29)$$

where the point O in Fig. 10 denotes the Big Bang point while the Hubble constant

$$H = \frac{1}{\Delta t} \quad (30)$$

denotes the inverse of the age of the Universe, which in this figure is described by the value Δt .

It follows that:

Galaxies move along their trajectories at a constant speed, similar to all other bodies in E4. The increase in the velocity of the galaxies as a function of distance was not the result of accelerating the galaxy motion but rather the result of the method of performing observations according to the E4 model.

Thus, the **introduction of the concept of dark energy**, which supposedly accelerated the expansion of the universe, **lost its justification**.

Notably, all interactions propagate along the wave ridges, which are perpendicular to the direction of wave motion—in this case, the motion of galaxies—in E4. Consequently, gravitational forces will not counteract the expansion of the universe in the t_i directions, since they act along directions perpendicular to the trajectories of the galaxies in E4, which are determined by the r_i directions—Fig. 7. Therefore, the only thing that gravitational forces can change is the direction along which certain galaxies expand in four-dimensional space. We perceive a change in the direction of a galaxy's trajectory as a change in its velocity, but this has nothing to do with the velocity at which galaxies move along their trajectories t_i in E4—that is, with the actual expansion velocity of the Universe.

We measure the speed of galaxies relative to Earth by analyzing the redshift of the radiation emitted by galaxies. The derivation in E4 of the formula for redshift based on the diagram shown in Figs. 1 and 7 is presented in the appendix in chapter 7.3 [Derivation of the formula for the redshift of the radiation of galaxies according to the proposed model of reality](#).

4.4 The problem with further modifying the theory

Changing the coordinate system from the one in which the phenomenon is observed to one that is independent of the observer is one way to increase the model's capacity. This is because greater freedom in defining the coordinate system automatically increases the possibilities for its development. A classic example of such an expansion of a model's capacity is the transition from the geocentric model—which described reality using a coordinate system bound to the Earth—to the heliocentric model, which employed a different coordinate system, which made it possible to expand the capacity of the reality model by introducing the freedom to choose a coordinate system independent of the observer's frame of reference on Earth. This made it possible, for example, to describe and plan interplanetary journeys.

In the E4 model, a change in the understanding of the coordinates that form reality has made it possible, for example, to eliminate singularities and predict an entirely new class of motions along trajectories for which singularities occur in Minkowski spacetime or which we are unable to describe using the dimensions of time and space that we are capable of observing (Fig. 6).

The change in the interpretation of spacetime dimensions—from dimensions that make up reality to directions within a certain objective reality described by distances that do not have a predetermined meaning as temporal or spatial distances—made it possible to unify a four-dimensional space in which motions are absolute with a three-dimensional subspace in which motions are relative, which allowed us to return to a model analogous to the original concept of the ether, which we assume here to be an elastic medium.

Describing space as an elastic medium, in turn, appears to be a promising way to overcome the difficulties in combining quantum mechanics with the theory of relativity, as well as a way to formulate a unified field theory.

For example, from the properties of a particle described as a wave (12), where the particle can be described as a compression or rarefaction of space rotating in the plane defined by the direction of the particle's motion in E4, compressions and rarefactions in space will cause particles with different types of deformation to attract one another and particles with the same types of deformation to repel one another—which explains the mechanism responsible for the existence of electric charges. In turn, the rotational motion of such a deformation—Fig. 5—has a certain angular momentum that resists a change in the wave's direction of motion, which in the new model determines the particle's relative velocity. Thus, this mechanism in turn defines the inertial forces and the origin of a particle's inertial mass. In this way, the new approach is very likely to create long-awaited unified field theory. Furthermore, the mechanisms responsible for electrical interactions, inertial mass, and likely gravitational mass become simple and easy to understand even without the use of complicated mathematics.

Moreover, the gravitational field is no longer simply a curvature of space of unknown origin. We can assume that, with a high degree of probability, gravitational interactions are caused by the angular momentum of the particles and the system's tendency to minimize the total angular momentum of the individual particles. It is still a kind of curvature of space, but it results from the superposition of the oscillations of individual particles; in other words, it is essentially the result of quantum effects. This already suggests a likely path toward unifying quantum theory with the theory of gravity. Of course, this still needs to be described more precisely in mathematical terms, but the idea and the direction in which further work should proceed seem to be taking shape more and more clearly.

The wave equation derived from the equations of quantum mechanics (12) describes the propagation of signals along the ridges of waves representing particles. This allows us to link the propagation speed of interactions to the coordinate systems of the particle-waves, exchanging those interactions, which, in turn, makes the speed of signal transmission independent of the observed relative motion of particles (the angle of inclination of the particles' trajectories), thereby justifying the constancy of the speed of light without the need to accept the ad hoc hypothesis, adopted in the 19th century, regarding the contraction of moving bodies. The constancy of the speed of light is explained by a specific mechanism and justifies relativistic effects; furthermore, treating quanta not as separate particles but as modulations of the wave that constitutes the particle additionally justifies their lack of mass.

Moreover, the idea of linking the speed of signal propagation to the frames of particles exchanging interactions turns out to be possible to verify experimentally in a relatively straightforward manner, both for this effect and for the assumptions underlying the Minkowski spacetime model—as described further in Section 5.3.

Thus, the new model provides a better understanding of the mechanisms underlying known phenomena and, above all, the creation of a unified field theory—a goal that previous models have struggled to achieve, facing difficulties that have thus far proved insurmountable.

4.5 Mach's principle

According to the new approach, the relative velocity of bodies is not related to any additional velocity relative to the medium, but it is a measure of the angle of inclination of the trajectory of the observed body to the trajectory of the observer. Thus, the relativity of the velocity of bodies in

the case of inertial motion, i.e., rectilinear trajectories in E4, consists of the relativity of the angle between the trajectories. By changing the angle of the observer's trajectory, we obtain different values of the velocity of the observed body; since there is no single distinguishing direction for trajectories, all the angles between trajectories are relative; hence, the velocities they describe are relative.

However, in the case of noninertial motion, which corresponds to a curved trajectory (Fig. 6), the radius of curvature of the trajectory of the observed body does not depend on the angle of inclination of the observer's trajectory. Hence, we can treat inertial motions as relative, whereas noninertial motions cannot be treated in the same way.

4.6 The impossibility of describing the universe in a coordinate system independent of the observer

According to the model presented here, although movements in three-dimensional space are relative, they are nevertheless the result of absolute motions along trajectories with varying inclinations in four-dimensional space, which in turn allows us to define a single common absolute four-dimensional reference frame in which the motions of all bodies can be described regardless of the observer's choice. The relativity of motion in three-dimensional space results from the relativity of the choice of the absolute direction of a body's motion in four-dimensional space and not from any additional velocity of bodies relative to the medium.

The model allows one to go to the description in the reference frame of a chosen observer at any time; however, in such a reference frame, there are certain limitations that do not exist in the absolute reference frame, such as the singularities discussed earlier. Thus, according to the E4 model, it is better to solve problems in Euclidean coordinates and then transform the result into the observer's spacetime coordinates. However, in regions where there are no singularities, it may be more convenient to use the existing description in the observer's spacetime coordinates.

4.7 The lack of a mechanism to justify the similarity of the dimensions of time and space

In Minkowski spacetime, one of the dimensions is designated the time dimension, and this is a property of the fourth dimension of reality. However, the similarity of the dimensions meant that once the distinction between spatial dimensions and time was hidden within the metric tensor, the equations of general relativity could be written in a form that treated all four dimensions equally. According to the currently accepted model, although the temporal dimension bears a strong resemblance to spatial dimensions, it differs slightly from them, and this difference is a property of spacetime, which describes physical reality.

Moreover, in the model proposed here, all four dimensions that make up reality are identical, but one of the directions is distinguished and interpreted as the time dimension of the observer's coordinate system—not because of the spatial properties but because of the motion of the wave describing the particle along that direction. Thus, the distances measured by a particle along its trajectory are interpreted by the observer as temporal distances, whereas the distances measured along the directions in which the observed bodies emit interactions are interpreted by the observer as spatial dimensions. Thus, according to the new model, the temporal or spatial nature of dimensions is not a property of physical reality but merely a property of the observational process. The same distance in E4 space may be a temporal distance for one

observer, while at the same time, it will be a spatial distance for another observer, which, I hope, convincingly explains the similarity of the observed spacetime dimensions.

5 A new concept of reality, new findings and new problems

This new approach to the problems of the structure of physical reality also involves the emergence of new conclusions and new interpretations of existing phenomena, as well as new, previously unknown problems.

5.1 The Dark Side of the Universe.

The mechanism of the expansion of the universe—Chapter 4.3—links the expansion of the universe to the properties of wave particles that move in E4 at a constant speed, radiating outward from the point that was the Big Bang. Thus, the universe is composed of particles that move along trajectories that, in every plane passing through the Big Bang point, fill an angle of 360° . However, the way we observe other celestial bodies or galaxies—that is, along a direction perpendicular to the trajectory of the observed body—limits our observations to only those galaxies moving along trajectories inclined at an angle of $\pm 90^\circ$ to Earth's trajectory, which in turn limits our observations to only half of the galaxies that fill the 360° angle. Thus, half of the Universe is beyond our observational reach. However, owing to the Earth's motion around the Sun, the boundaries of the observable universe shift slightly, which should cause the most distant galaxies to disappear from view at certain times of the year. This problem is illustrated in Fig. 8.

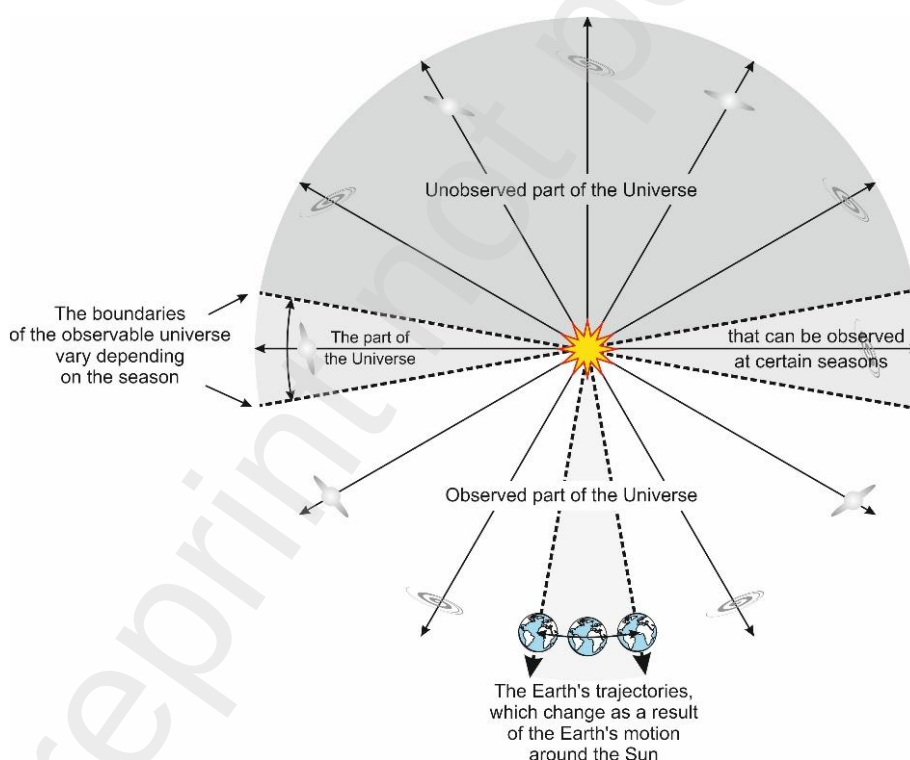


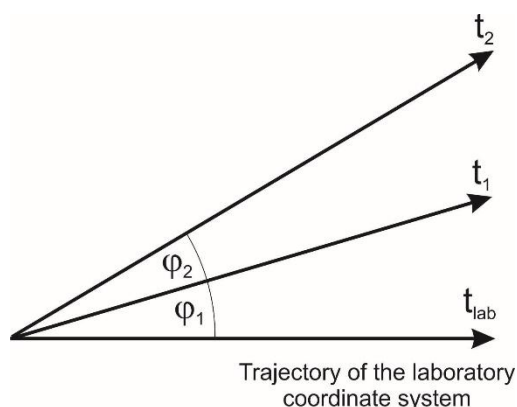
Fig. 8 The dark side of the universe includes galaxies moving away from the Big Bang along trajectories inclined at an angle greater than or equal to 90° . Changes in Earth's trajectory resulting from its orbit around the Sun alter the boundaries of the observable universe. Therefore, the most distant visible galaxies should appear only at certain times of the year.

Calculations of the Earth's orbital speed around the Sun indicate that such an effect should be observable for galaxies with a value of redshift greater than 100, which is currently well beyond the range of astronomical observations, where the maximum redshift value is approximately 20.

5.2 New rule for composing velocities and coordinate transformation

5.2.1 New rule for composing velocities

In the E4 model, the relative velocity is the angle of inclination of the trajectory, and the relative velocity is equal to the sine of the angle between the trajectories of the bodies. Therefore, the composition of velocities here involves adding angles between trajectories; this is illustrated in Fig. 9.



*Fig. 9 Composition of velocities on the E4 model. The bodies t_1 (**body 1**) and t_2 (**body 2**) move relative to the laboratory system t_{lab} and relative to each other. The measure of relative motion is the angle between the trajectories related to the observed velocity by formula $V = \sin\varphi$ (6). The motion of the body t_1 relative to the laboratory system is determined by the angle φ_1 , while the relative motion of the bodies t_1 and t_2 is determined by the angle φ_2 . Thus, the measure of the motion of the body t_2 relative to the laboratory system is the angle $\varphi_1 + \varphi_2$ which corresponds to the velocity $V = \sin(\varphi_1 + \varphi_2)$.*

In this case, if the velocity of body 1 in a laboratory system is:

$$v_1 = \sin\varphi_1 \quad (31)$$

The velocity of body 2 relative to body 1 is as follows:

$$v_2 = \sin\varphi_2 \quad (32)$$

Then, according to Figure 9, the resultant velocity of body 2 relative to the laboratory system should be equal to the sine of the sum of the angles φ_1 and φ_2 and equal to the following equation:

$$V = \sin(\varphi_1 + \varphi_2) = \sin\varphi_1 \cos\varphi_2 + \sin\varphi_2 \cos\varphi_1 = v_1 \sqrt{1 - v_2^2} + v_2 \sqrt{1 - v_1^2} \quad (33)$$

As shown in the diagram in Fig. 10, the speed of light will not be exceeded, while the speed interpreted as the speed of light is reached at one of the points in the diagram after two specific values of speed are composed, each of which is less than the speed of the light.

However, once the velocity reaches $V = 1$, it begins to decrease. Nevertheless, as shown in Fig. 6, the concept of velocity does not apply in this range of angles ($\varphi \geq \frac{\pi}{2}$) since it is valid only for smaller angles. It is possible to analyze motions along trajectories with angles $\varphi \geq \frac{\pi}{2}$ in E4, but only by using the concept of a trajectory. Interestingly, R.F.J. Van Linden [40]

independently derived exactly the same principle of velocity composition (33) using a Euclidean spacetime model constructed from spacetime dimensions.

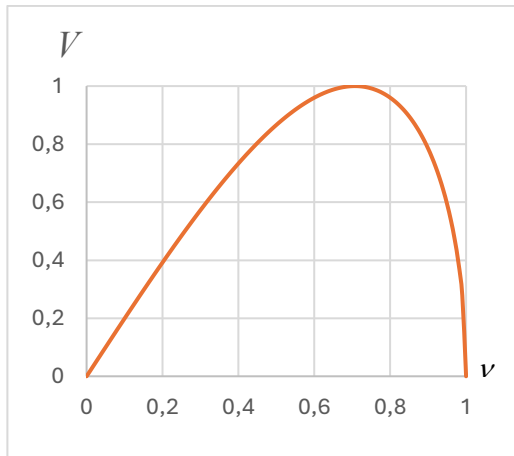


Fig. 10 Result of composing two identical velocities according to model E4. The velocity composing principle does not allow to achieve a velocity greater than unity (the speed of light), but for certain values of the combined velocities v , the resultant velocity can reach the value of $V=1$, which is impossible in the case of composing velocities in the Minkowski spacetime model.

The new principle of velocity composition (33) does not preclude the possibility that bodies may follow a trajectory inclined to the observer's trajectory at an angle of $\varphi \geq \frac{\pi}{2}$. However, while such an effect could not be observed in the case shown in Fig. 6, in the case of particle decay, a product moving along such a trajectory would not interact with the detectors of the laboratory setup, so we would have a mass defect in such a reaction. This is similar to the description of a reaction involving neutrino emission, although drawing such conclusions requires much more detailed analysis.

In addition, the new principle of velocity composition makes it possible to achieve a relative particle velocity of $V=1$, for which a singularity occurs in the spacetime description. This should, for example, result in certain differences between particle measurements taken in a collider and those taken in systems with a stationary target. However, as I show in [57], this requires a very high degree of energy homogeneity in the collider's beams, so it seems that the only thing that can be demonstrated is an increase in statistical errors in these energy regions.

5.2.2 The new coordinate transformation

In accordance with the model presented here, a mechanism is introduced that links the transmission of interactions to the coordinate systems of the particles exchanging those interactions. This explains why the speed of light remains constant regardless of the relative velocity of the bodies. As a result, it is no longer necessary to assume Lorentzian contraction to justify the constancy of the speed of light. Thus, in the E4 model, the justification for the constancy of the speed of light does not require the deformation of the spacetime described by the spacetime interval equation. However, the space-time interval equation is still applied here but only in a form restricted to the problem of describing the mutual observation of bodies (34), whereas the problem of coordinate transformation when changing observers is derived here from a comparison of coordinates in the reference frames of two observers observing the same body. The problem is illustrated in Fig. 11a, where observers t and t' observe body t_0 .

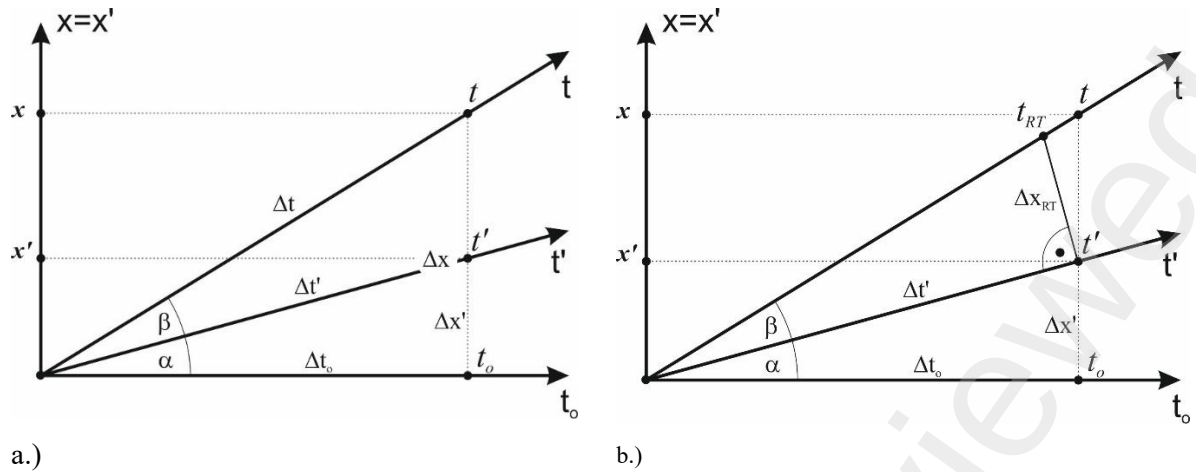


Fig. 11 Observers x, t and x', t' observe the body t_0 . In Figure a, the observers do not observe one another. In Fig. b, observer x, t also observes the motion of observer x', t' . The observation is conducted along directions perpendicular to the trajectory of the observed body—here, the observed distance between the observers is defined by the segment Δx_{RT} , whereas observer x, t observes that if time t' has elapsed in the reference frame of observer x', t' then time t_{RT} has elapsed in his own reference frame.

In this case, as presented in Fig. 11a, the space-time interval conservation equation takes the following form:

$$\Delta s^2 = \Delta t_0^2 = \Delta t^2 - \Delta x^2 = \Delta t'^2 - \Delta x'^2 \quad (34)$$

However, this simplified equation for the conservation of the spacetime interval in E4 describes only the dependence on the times elapsed in the x, t i x', t' systems—that is, the times Δt and $\Delta t'$ relative to the observed body t_0 , i.e., relative to the time Δt_0 . Thus, equation (34) allows us to determine the relationships between the coordinates of observers in E4, but it does not allow us to determine the relationships between the spacetime frames of observers moving along trajectories t and t' . To determine this relationship, we must supplement equation (34) with the equation describing the observation of body x', t' from the frame of body x, t —that is, with the equation resulting from Fig. 11b.

$$\Delta s_{RT}^2 = \Delta t'^2 = \Delta t_{RT}^2 - \Delta x_{RT}^2 \quad (35)$$

Thus, equations (34) and (35), which are needed to determine the spacetime coordinate transformation, are now derived from Fig. 11b rather than from Fig. 11a.

Let the relative velocities of observers x, t and x', t' be denoted as follows:

$$V = \sin \beta \quad (36)$$

The body's velocity t_0 is measured in the observer's reference frame x, t as follows:

$$v = \sin(\alpha + \beta) \quad (37)$$

We obtain the equation for the coordinate transformation in the form:

$$t = \frac{t'}{\sqrt{1-V^2}} + \frac{x'V}{\sqrt{1-V^2}\sqrt{1-v^2}} \quad (38)$$

$$x = x' + \frac{t'V}{\sqrt{1-v^2}} \quad (39)$$

Note that, unlike the Lorentz transform, which defines the relationship between observers' coordinates solely in terms of their relative velocity, denoted here by V in the new transformation

(38,39), the coordinates also depend on the velocity of the observed body relative to one of the observers - v . This is because, according to the E4 model, the Lorentz transformation is incorrect—more so can be found in [51, 61]. Furthermore, as can easily be verified, the solution to the spacetime interval conservation equation is a pair of functions of the form:

$$ct = \frac{ct'd + x'e}{\sqrt{d^2 - e^2}} \quad (40)$$

$$x = \frac{x'd + ct'e}{\sqrt{d^2 - e^2}} \quad (41)$$

Since substituting formulas (40) and (41) into the spacetime interval equation causes the factors d and e to be shortened, whatever we substitute in their place, the formulas always satisfy the spacetime interval conservation equation. When $d = 1$ and $e = \frac{v}{c}$, this equation describes the well-known Lorentz transformation however, in this context, any “derivations” of the Lorentz transformation from the spacetime interval equation, such for example, as in [71], are nothing more than a proof presented in a different way, demonstrating that the Lorentz transformation satisfies the spacetime interval conservation equation. The Lorentz transformation is, in fact, one of an infinite number of such possible transformations.

A more detailed discussion of the problems associated with the Lorentz transformation can be found in [51, 61].

5.3 When light travels faster than light

The relationship between the propagation of the quanta—which is now treated as a disturbance of the wave that constitutes the particle—and the particle coordinate system, rather than the absolute coordinate system describing E4 space (described in detail in 7.1 [Mutual observation of bodies](#)), justifies the constancy of the speed of light without the need to assume Lorentz contraction, as described in the previous subsection. However, the accepted rules imply that the trajectory of a signal transmitted in a manner associated with the coordinate systems of particles exchanging the signal is not fixed relative to space but is determined by the trajectories of the transmitter and receiver in E4. While in the case of two bodies, the problem allows for a precise correlation between the distance between the transmitter and the receiver and the signal propagation time, the introduction of additional bodies—such as a relativistic signal relay—breaks the propagation problem down into two distinct problems—the transmission of the signal from the transmitter to the relay and then the transmission of the signal from the relay to the receiver. As a result, the signal path in E4 becomes a broken line—one for signal transmission between the transmitter and the repeater and the second between the repeater and the receiver; consequently, when an additional repeater is used, as one of the commenters on my YouTube channel (YouTube.com/euclideanreality) pointed out to me [72], the signal may reach the receiver in less time than it would happen for a signal sent directly to arrive.

Since, in the E4 model, a quantum is not treated as an independent particle but rather as a disturbance traveling along the ridges of waves carrying the signal, with a phase velocity equal to one in the frame of each body, therefore, we cannot speak of the speed of a quantum's propagation relative to other bodies that are not “carriers” of the disturbance that the quantum now represents because only bodies exchanging signals can detect the quantum's movement. Therefore, the shorter signal transmission time does not violate the condition that the speed of light is constant.

The case of signal transmission both directly between the receiver and the transmitter and via a relativistic relay for bodies that were launched simultaneously from the same point in four-dimensional space is illustrated in Fig. 12.

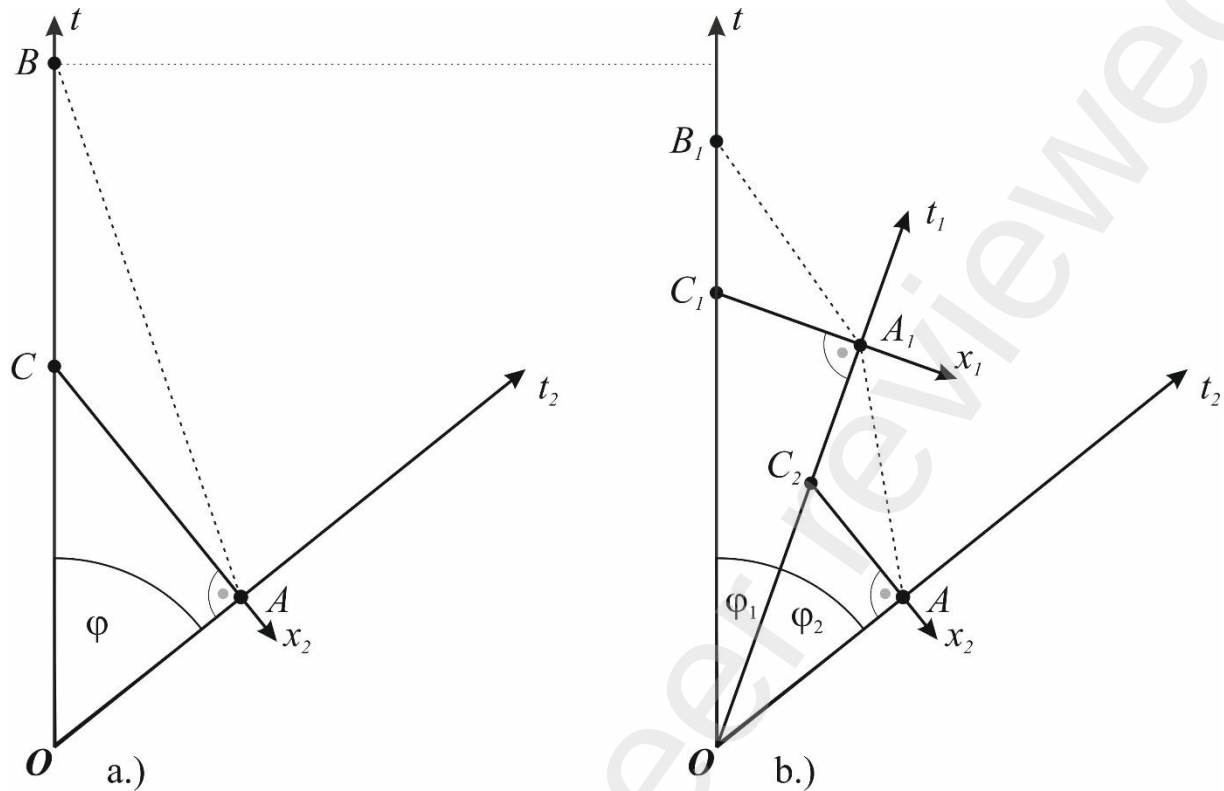


Fig. 12. Signal transmission between a transmitter moving along trajectory t_2 in E_4 and a receiver moving along trajectory t . The signal relay moves along a trajectory t_1 . The case shown in this figure concerns bodies that were located at point O at time $T = 0$. Figure a.) shows the direct transmission of a signal sent by a transmitter at point A along its trajectory and received at point B along the receiver's trajectory. Figure b.) shows signal transmission via a relay moving along a trajectory t_1 . It is clear that when a signal is transmitted via a relay, it reaches the receiver faster than in the case of direct transmission. This is due to the change in the signal path caused by the addition of the relay. The constancy of the speed of light ensures that the segments representing time and spatial distances, respectively, in the coordinate systems of the transmitter, receiver, and relay—that is, segments CA and CB in Fig. a—are equal, as are segments C_2A and C_2A_1 and C_1A_1 and C_1B_1 in Fig. b.).

The exact reduction in the signal propagation time from the transmitter to the receiver when a relay is used can be determined using the simple equations shown in Figure 12.

The following relationships can be derived from Fig. 12a:

$$CA = \Delta x_2, CB = \Delta t, \frac{\Delta x_2}{\Delta t} = 1, OA = \Delta t_2 \quad (42)$$

From this, we can calculate the time it takes for the signal sent by the transmitter at point A to reach the receiver at point B :

$$\Delta t = OB = OC + CB = OC + CA = \frac{OA}{\cos \varphi} + OA \operatorname{tg} \varphi = \Delta t_2 \left(\frac{1+V}{\sqrt{1-V^2}} \right) = \Delta t_2 \sqrt{\frac{1+V}{1-V}} \quad (43)$$

Applying this reasoning to describe the transmission of a signal from the transmitter to the relay and then from the relay to the receiver, we see the following:

As shown in Figure 12b,

$$V_1 = \sin\phi_1, V_2 = \sin\phi_2, C_2A = \Delta x_2, C_1A_1 = \Delta x_1 \quad (44)$$

Therefore, the time it takes for the signal to travel from the transmitter to the receiver is as follows:

$$\Delta t = OB_1 = OC_1 + C_1B_1 = OC_1 + C_1A_1 = \Delta t_1 \sqrt{\frac{1+V_1}{1-V_1}} \quad (45)$$

The time it takes for the signal to travel from the transmitter to the relay is as follows:

$$\Delta t_1 = OA_1 = OC_2 + C_2A_1 = OC_2 + C_2A = \Delta t_2 \sqrt{\frac{1+V_2}{1-V_2}} \quad (46)$$

From this, we can see that the total signal transmission time from the transmitter through the relay to the receiver is (Fig. 12b):

$$\Delta t = \Delta t_1 \sqrt{\frac{1+V_1}{1-V_1}} = \Delta t_2 \sqrt{\frac{1+V_2}{1-V_2}} \sqrt{\frac{1+V_1}{1-V_1}} \quad (47)$$

As shown in Figures 12a and 12b, the signal is transmitted by the transmitter at point A along the transmitter's path and reaches the receiver at points B and B₁. In Figure 12a, the signal is transmitted directly from the transmitter to the receiver. In Fig. 12b, the transmitter sends a signal to the relay, where it arrives at point C₁, and is then sent to the receiver, where it arrives at point B₁. The constancy of the speed of light ensures that spatial and temporal segments are equal along the directions in E₄, which are interpreted as the axes of the reference frames of the receiver, the relay, and the transmitter—Fig. 12. The spatial axis of the body currently receiving the signal is perpendicular to the trajectory of the body transmitting the signal—thus, as shown in Fig. 12a, the observer records the signal received directly from the transmitter in the t, x_2 coordinate system, while the signal received from the relay—Fig. 12b—is recorded in the t, x_1 coordinate system. In turn, the relay describes the signal received from the transmitter in a coordinate system defined by the directions t_1, x_2 . It can be said that the reduction in signal propagation time resulting from a change in the propagation path of disturbances caused by the introduction of a relay is comparable to the effect of a change in the time it takes for light to reach its destination due to the refraction of light at the boundary between different media.

The situation in which all three components—the transmitter, receiver, and relay—were initially located at the same point is shown in Figure 12. However, a similar phenomenon should also occur when the trajectories of the transmitter and receiver are parallel or nearly parallel. In this case, using a relay also alters the signal path and changes the time it takes for the signal to travel from the transmitter to the receiver. For example, if we were to send a signal to the moon or Mars via two different paths—directly and via a relay on a spacecraft traveling to that celestial body—the received signals should, in theory, be separated by a time difference resulting solely from the distance traveled by the signal—whether directly or via a relay in the rocket. Moreover, if the signal propagation model proposed here—which relies on ensuring the constancy of the speed of light not through Lorentz contraction but by linking the signal's motion to the coordinate systems of the bodies involved in signal transmission—is correct, then the signal from the transmitter should reach its destination faster (or later—depending on the direction of the relay's movement) than that predicted by calculations based on the theory of relativity. The estimated

time difference for sending a signal to Mars is a few milliseconds, whereas for sending a signal to the moon, it is a few microseconds. Of course, exact values can be calculated using precise distances and the rocket's speed, but it is clear that these are measurable quantities. Since Figure 12 uses the new principle of velocity addition described in more detail in the previous chapter, a positive result from such an experiment should also confirm the validity of the new principle of velocity addition and the new form of the coordinate transformation.

Thus, the example mentioned above may be the easiest way to test the new understanding of physical reality proposed in this work.

5.4 The Problem of the Concept of Motion.

According to the theory of relativity, the motion of bodies is described by a world line, which shows the positions of the body in spacetime. If the world lines of two bodies in spacetime intersect at a common point, it means that the bodies will meet at that point.

The E4 model assumes that bodies move along certain trajectories in four-dimensional reality. A trajectory differs from a world line in that it represents only the paths of bodies in space but does not specify at which point in time a body is located at a given point along the path. To determine the position of bodies along a trajectory, it is also necessary to specify time, which is not a fifth dimension here but merely a parameter that orders events and is common to all bodies. Therefore, for the bodies to meet, they must be at a point with identical coordinates in E4 space and at the same time. Thus, it is no longer possible to represent motion in E4 on a plane—as was the case in the theory of relativity—but requires at least a three-dimensional representation, where two axes define trajectories in a plane within four-dimensional space, and the third axis determines the value of the parameter T —that is, the position on the trajectory for all bodies. An example of such a presentation is shown in the next subsection, where we discuss the motion of massive bodies and light quanta.

5.4.1 The motion of massive bodies and quanta

An example of the motion of bodies and quanta in E4 is shown in Figure 13.

The diagram illustrating the problem in E4 space shows Earth and a hypothetical planet. The distance between the Earth and the planet is constant. A rocket is launched from Earth toward the planet at a speed equal to $V = 1$ —which is not permitted in the theory of relativity; however, in the E4 model—as shown in Fig. 6—it is possible to describe motion along such a trajectory. At the same time, a laser beam is being sent toward the planet.

The trajectories in the plane within the E4 space are shown in Figure 13a. The rocket's trajectory—and, at the same time, the time axis of the observer's coordinate system inside the rocket—is perpendicular to the trajectory (and, at the same time, the time axis) of the Earth and the planet—which corresponds to the rocket's velocity being equal to one. The trajectory of the laser beam is inclined relative to the trajectories of the Earth and the planet by an angle of $\varphi = \frac{\pi}{4}$ because the motion of the quanta is the superposition of the disturbance motion along the direction of motion of the Earth and the planet in E4 and the motion perpendicular to the planet's trajectory in the planet's coordinate system, also at a speed of $V = 1$ (as described in detail in the appendix [Mutual observation of bodies](#)). Looking only at Fig. 13a, we see that in the Earth's coordinate system, the time elapsed in the rocket is zero; however, because the rocket is moving along its trajectory, an observer inside the rocket sees that his watch also shows the length of the trajectory traveled by the rocket. However, if a rocket and a laser beam—moving in the frame of an observer on Earth at a speed that is both equal in magnitude and equal to unity—

will they reach the planet at the same time? This is illustrated in Fig. 13b, where we add a dimension representing the absolute time T that describes the position of all these objects along their trajectories.

We see that after time $t = T_p$, the Earth and the planet will have moved along their trajectories from point $t_0 = 0$ to point $t = T_p$. At that time, the rocket will move in a direction perpendicular to the trajectories of the Earth and the planet to the point $r = T_p$; that is, the rocket's trajectory will indeed intersect the planet's trajectory, but at the point where the planet was at time $t_0 = 0$. Therefore, the hypothetical rocket traveling at the speed of light would not reach the planet. In turn, the motion of a light ray is the combination of two motions: motion along the trajectories of the Earth and the planet t at a speed of $V = 1$ and motion in a direction perpendicular to these trajectories— r —(resulting from the phase velocity of the wave along its ridge) at a speed of $V = 1$. Thus, in this specific case, the speed of the light ray relative to E4 will be $V = \sqrt{2}$; however, an observer on Earth and on the planet sees, in their reference frame, the light ray moving at a speed of $V = \frac{\Delta r}{\Delta t} = 1$.

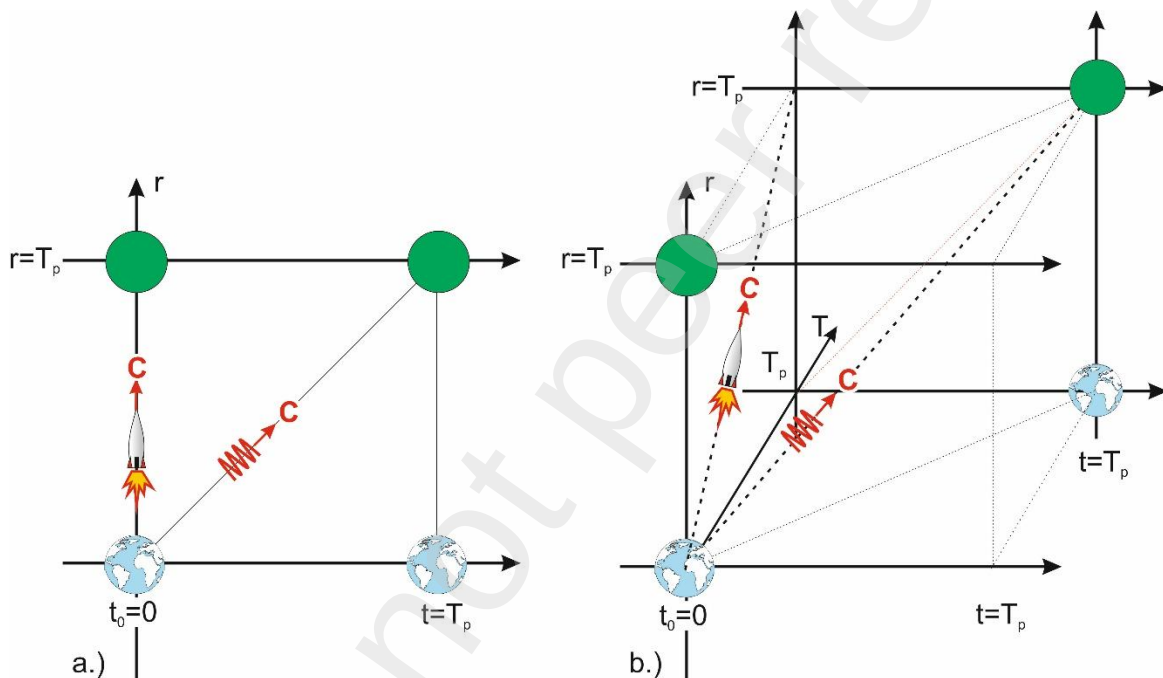


Fig. 13. A diagram illustrating the transmission of a light signal from Earth and a hypothetical rocket traveling at the speed of light toward an example planet. The trajectories of the Earth, planet, and laser beam in the E4 plane are shown in Fig. 13a. The absolute time axis is shown in Figure 13b. All the bodies and the laser beam move in a plane whose value is equal to the absolute time common to all the bodies. This plane defines the current positions of the objects along their trajectories and moves in the direction determined by the absolute time axis T .

Thus, it is clear that the motions of massive bodies and light are entirely different phenomena. Applying the concept of velocity—measured in the observer's coordinate system—to both of these phenomena leads us to treat them as identical motions in space, even though they are, in fact, entirely different phenomena.

Figure 13 shows a special case of the motion of bodies. However, if a rocket traveling along the trajectory shown in Fig. 13 does not hit the planet, will it reach the planet if it travels along a

different trajectory? No, not because the rocket's velocity must have a component equal to one that is parallel to the trajectory of the planet and Earth but because the rocket's trajectory is inclined at a certain angle to the trajectory of the Earth and the planet, which are also moving at a velocity of $V=1$, so it must have an additional perpendicular velocity component that allows it to "catch up" the planet. In other words, the length of a rocket's trajectory during interplanetary travel is greater than the length of the Earth's or a planet's trajectory; assuming that the planet and the rocket are moving at constant speeds along their own trajectories, this makes it impossible for the rocket and the planet to intersect, even when their trajectories intersect. The same applies to a rocket that, as it moves away from Earth and then returns, follows a longer trajectory than Earth itself does. The problem is much more serious because, according to the current model—which assumes that objects move at a constant speed along their trajectories—even a simple commute to work would be impossible. In addition, yet we travel to the moon, to Mars, and commute to work every day.

How can we address this problem?

5.4.2 Noninertial motions

Thus far, we have considered only inertial motion, that is, motion along straight-line trajectories. However, the trajectories of bodies are in fact of different lengths, and therefore, mechanisms must be introduced that ensure that, despite moving along trajectories of different lengths, systems of bodies composed of elements moving relative to one another along trajectories of different lengths—such as planetary systems, atoms, chemical compounds, or people moving relative to the Earth's surface—remain constant and stable.

The model presented here assumes that the trajectories of bodies in four-dimensional space are arbitrary, which rules out the idea of treating three-dimensional space as a three-dimensional layer propagating within four-dimensional space, as proposed by some models describing Euclidean reality [8,9]. The model adopted in this work, which defines spatial dimensions not as fixed directions in E4 space but as directions perpendicular to the trajectories of bodies in E4—and thus, in practice, directions that can be oriented in various ways within E4 space—yields two significant properties of the new model, namely:

On the one hand, it allows for what is probably the simplest possible description of reality by adopting a model of reality as a four-dimensional elastic medium in which waves exist that move at constant speeds along arbitrarily oriented trajectories and exchange interactions along their three-dimensional ridges. The simple structure of the model makes it possible to immediately link the concept of absolute motion with relative motion to justify the constancy of the speed of light without having to assume that bodies in motion are deformed and to combine the wave description of bodies with relativistic effects or a simple derivation of the mechanism underlying both electromagnetic and, presumably, gravitational interactions (currently, we have only a description of inertial mass) and many other interesting conclusions.

On the other hand, by assuming that the directions of motion of bodies—and thus the directions interpreted as temporal and spatial in E4—are arbitrary, we rule out the possibility of defining three-dimensional space as a layer within four-dimensional space. Thus, we are faced with the problem of defining an alternative mechanism that ensures that wave systems moving along trajectories of varying lengths in E4—such as particles in an atom or planetary system—retain their form; in other words, all these particles are constantly in a hyperplane with a constant value of absolute time T .

There are two possible ways to solve this problem. One approach might be to assign the nature of a fifth dimension to absolute time and to define reality as a four-dimensional layer moving along the fifth dimension, as shown in Fig. 13b, where all the bodies move along the time dimension in a plane with a constant time value T common to all the bodies. However, this adds an unnecessary level of complexity to the model, which may not be necessary.

The second method involves temporarily adopting a mathematical description based on the assumption that all bodies move in such a way as to preserve the absolute simultaneity described in the appendix in Chapter 7.1.1 [Simultaneous events](#). At this point, identifying the mechanism responsible for this behavior of the bodies is not possible. It can be surmised that this is related to a change in the direction of wave motion in E4, which must induce different vibration frequencies along its ridge when the direction of motion changes. However, these are merely speculations at this stage, and the mechanism requires further clarification. As shown in section 7.2 [Mechanism to change the time](#), a mathematical trick is currently used that allows us only to describe the trajectory such that the motion of noninertial particles along a longer trajectory connecting two points in E4 corresponds to a shorter elapsed time in the noninertial particle's frame (relative to the time for a straight-line trajectory). Incidentally, the model clarifies the issue of time dilation by describing dilations in an inertial frame not as an actual slowing down of time but merely as the result of an observation that is symmetrical for both observers, and it is only the disruption of this symmetry caused by a noninertial episode in the motion of one of the observers that causes the previously mutually symmetric observed mutual time dilation to become an actual time dilation in the reference frame experiencing the noninertial episode. I have written about this issue in my earlier works when discussing the Hafele and Keating experiment [45,46].

Thus, while there are works that propose a model of physical reality similar to the one presented in this and my previous works [47–66], namely, a four-dimensional Euclidean space composed of four identical dimensions, with particles moving within it and absolute time possessing the characteristics of a parameter—e.g., [9], in the work presented here, we have an extremely simple model of reality that immediately explains many previously unexplained phenomena, but a new problem arises regarding the description of noninertial motions. Moreover, competing concepts generally retain the interpretation of spatial dimensions as actual dimensions that constitute reality, with all its consequences—such as the deformation of dimensions in motion and the preservation of many elements of the theory of relativity, which can be described only through mathematics, whereas some of these models allow space to be represented as a layer in four-dimensional Euclidean space, which provides a simple solution to the problems of noninertial motion. Such a model is probably easier to grasp than the one presented in this paper, as it contains many familiar elements of the theory of relativity however the interpretation of the dimensions of Euclidean space presented in this paper and in previous ones [47–66] allows for a significant simplification of the model and the solution of many problems previously considered unsolvable, however, at the cost of major changes in the interpretation of the structure of reality, consisting mainly of a change in the interpretation of spatial dimensions—from dimensions constituting actual reality to directions dependent on the trajectory of the currently observed body. Thus, according to the model presented here, the spatial dimensions we observe are not a property of reality but merely a property of the process of observing objects.

6 Conclusions

In this paper, I have presented an approach to the problems of the structure of reality that is entirely different from the one currently accepted in science. In the early 20th century, while Lorentz was working on a model of the ether that could reconcile the relative motion of bodies with the constant speed of light, Einstein published a paper in which he stated that the existence of the ether is not necessary to explain relativistic phenomena. Instead of trying to reconcile the two speeds—constant and relative—which seemed irreconcilable within the then-prevailing model of three-dimensional space and time, Einstein declared that the speed of light is constant because it is a fundamental property of reality and that the ether is unnecessary. Instead of continuing to investigate the mechanical properties of the ether to explain the phenomenon of the constancy of the speed of light, this was replaced by postulates and mathematics, which yielded rapid successes; in fact, for more than 100 years, no one has seriously pursued a mechanical model of the ether, which, in the face of sophisticated mathematics, seemed obsolete. However, the acceptance in the late 1920s of new features of reality—such as the belief in its four-dimensional nature and the wave nature of matter—which were unknown to Einstein when he developed the theory of relativity, made it possible to view the problem of the existence of a medium serving as an absolute frame of reference from a new perspective. Moreover, the problems emerging in quantum mechanics due to the absence of a medium, which the ether was supposed to be, suggested the need for a new approach to physics.

In this article, I return to the research on the nature of the ether, which was interrupted by Einstein's 1905 publication, and present a proposal for mechanisms that explain the phenomena previously accepted as the basis for current models—that is, answers to the following questions:

1. Why is the speed of light constant?
2. Why are the dimensions of spacetime different?
3. What causes both wave-like and particle-like behavior?

While it proved impossible to answer these questions using Minkowski's spacetime model, a radical reconstruction of the model of reality made it possible to explain these issues using simple concepts.

6.1 What was done in this article

First, the underlying framework of the model was modified by replacing Minkowski's four-dimensional spacetime with four-dimensional Euclidean space, supplemented by the fact that particles move within this space—in the case of motion along straight-line trajectories, this is motion at a constant speed. It should be emphasized here that this does not mean that the concept of Minkowski spacetime is incorrect. Minkowski spacetime describes the observed picture of reality, whereas the E4 model aims to describe objective reality, which exists independently of the observer and the process of observation. The motion of bodies in E4 is defined relative to absolute time, which is not an additional dimension here but a parameter common to all particles. The absolute motion of particles along different trajectories causes these particles, even when they move in E4 at identical speeds, they observe other particles as moving relative to one another, whereby the observed relative motion of the particles does not result from any additional motion of the particles relative to E4 but is the result of motion in E4 along trajectories inclined at various angles to one another. As a result, even though all dimensions are identical in E4, particles perceive their trajectory in E4 as a time axis, and

because of the way observations are made, directions perpendicular to the trajectories of the observed particles are interpreted by observers as the spatial dimensions of their coordinate systems. Owing to the freedom of choice regarding trajectories, when observing each body, we interpret different directions of E4 as spatial dimensions. Consequently, according to the model presented, what we perceive as space is not a static entity but a dynamic, observable structure formed by the superposition of various directions of interaction propagation, which depend on the various directions of the trajectories of bodies in E4. In practice, as we observe our surroundings, we construct our picture of reality from many independent observations of individual objects. However, when we observe different bodies, we interpret different directions in E4 as identical spatial dimensions in the observer's frame of reference perceived by the observer as relativistic effects. **Thus, it is not the deformations of the dimensions of moving bodies but the fact that, when observing these bodies, we interpret the various directions in E4 as our spatial dimensions that are the source of relativistic effects.**

Another feature of the new model is the ability to describe bodies directly as waves that are disturbances in space. This has become possible because of the assumption of a constant velocity of motion for bodies in E4. A wave in three-dimensional space is a vortex in four-dimensional space. Representing a wave as a deformation of space in E4 rotating in a plane defined by the direction of the wave-body's motion in E4 allows us to represent the body as a rotating compression or rarefaction of space. This representation of the body—as a wave—directly implies the properties of attraction between waves with opposite types of spatial deformation (compression and rarefaction) and the repulsion of waves with the same types of deformation—which can be interpreted as electrical interactions—and simultaneously, the angular momentum resulting from the rotational deformation in E4 counteracts changes in the direction of the body-wave's trajectory (interpreted as a change in velocity), indicating the effect of inertial mass. At the same time, waves moving in E4 at a speed of $V=1$ observe waves moving along other trajectories (inclined at various angles to the observer's trajectory) as wave particles move relative to them at a relative speed. Therefore, waves that require a constant velocity relative to the medium to be described can simultaneously be described—that is, they can be observed from one another's perspective—as moving at different relative velocities, which does not alter the value of the wave's velocity relative to the medium.

The final element that completes the entire model links the transmission of information between bodies to systems of bodies that exchange information. As a result, the signal transmission speed within each body's system is $V=1$, while signal transmission first involves the propagation of a signal (a disturbance in the wave's amplitude) along the ridge of the wave sending the signal, and after the signal passes to the ridge of the wave receiving the signal, the signal propagates along the ridge of that wave. Signal propagation defined in this way ensures a constant signal propagation speed that is independent of the angle between the trajectories—that is, of the relative velocity of the interacting bodies.

As you can see, the model's structure is surprisingly simple, and all of its components come together to form a coherent whole, naturally describing both relativistic and quantum effects within a single model.

6.2 What remains to be done or clarified

This chapter presents fundamental changes in the model of reality that reveal the mechanism underlying these phenomena but does not yet describe them quantitatively. Many properties are given only as symbols that have not yet been defined numerically. However, the aim of this study was precisely to identify a coherent mechanism that explains properties whose values have

been known for some time but whose values have been accepted without an understanding of the mechanisms responsible for them. Given our current understanding of these mechanisms, we will most likely be able to determine the values of the new coefficients in the near future; however, this is a topic for future research, given the need to quantitatively describe specific mechanisms and the associated need to resolve new issues concerning the quantitative properties associated with this complex type of spatial deformation.

Since mathematics is a very versatile tool and allows us to draw correct conclusions on the basis of incorrect assumptions—as was the case, for example, with Ptolemy's epicyclic models—the introduction of mathematics should take place only at the stage when we have a complete model of the phenomena responsible for the observed nature of reality. It appears that, at this stage of the work, the fundamental properties of the model responsible for relativistic and quantum phenomena, as well as the source of the forces of electrical interaction, inertia, and—in all likelihood—gravity, already provide a sufficient basis for introducing the mathematics; however, that is a topic for several future papers.

Therefore, we have a number of problems here that can be solved using mathematics and the methods described in this paper.

Let us begin with the wave equation described by equation (12), the equation relating the energy of spatial deformation to the amplitude (15), and the postulated equation for the energy of wave deformation as a function of time (17). Equation (12) includes the amplitude of the oscillation, i.e., the magnitude of the spatial deformation. The first step should be to determine the spring constant and amplitude such that the interaction energy described by equation (17) or (24) results in an interaction energy identical to that of electric charges. Then, one could determine the spatial deformations corresponding to the electric charge. Moreover, the change in the wave distances corresponding to the charges should result in a change in the energy of the particle system, as in the case of the motion of charges in an electric field.

According to the model proposed here, the E4 model does not include concepts such as electric field or charge. These are merely properties observed in our reality; however, according to the E4 model, such concepts do not exist. The electric field, potential, field strength, and electric charge that we observe are the results of the interaction of space deformations aimed at reducing the total energy of the system of bodies. Another consequence is that the rotation of the three-dimensional deformation of space, representing a particle, in four-dimensional space E4 is associated with inertial mass and, consequently, gravitational mass. The angular momentum resulting from this motion is described in our space by Planck's constant; however, the relationship between the energy associated with the deformation of space and its rotation in E4 and this angular momentum remains to be explained. Ultimately, the spatial spring constant from equation (15) should replace both the electric charge—in terms of the spatial deformation—and Planck's constant—in terms of the rotation of that deformation—but this requires solving new problems that arise when we have only a four-dimensional medium and its deformations at our disposal.

Another problem awaiting a solution is the issue of changes in the velocity of bodies. In E4, this refers to the rotation of a wave—but when waves extend to infinity in E4, one might ask the following question: How does such a wave behave when it changes direction? I attempt to describe this in Appendix 7.2, "Mechanism to change the time," although these are only initial attempts for now. Moreover, they indirectly suggest the possibility of linking time dilation to the phenomenon of quantum entanglement, although this may still be too far-fetched an analogy.

The fact that signal propagation is linked to systems of particle exchange interactions gives rise to another class of problems concerning, for example, the conditions for signal transmission between wave ridges or signal propagation under noninertial motion.

In summary, the current form of the model serves as a framework that defines common mechanisms allowing all phenomena—previously described by different models—to be described by a single model; however, this framework must be filled in by calculating new, common fundamental constants primary relative to those currently in use, such as that postulated in formula (15) the spring coefficient, or by conducting an analysis of possible four-dimensional vibrations of E4 space and its likely resonance states. Thus, in this sense, the proposed model does not provide a solution to existing problems but rather marks the beginning of an entirely new way of perceiving reality, one that should be evaluated and either further developed or modified as new problems arise.

Finally, a few words are used to verify the model's predictions. Two proposed experimental tests seem feasible to carry out either now or in the near future. While the predicted effect of the visibility of the most distant galaxies at certain times of the year—Chapter 5.1, “The Dark Side of the Universe”—is currently beyond the reach of our observations (it requires observations of galaxies with a redshift coefficient of $z > 100$, which is currently unattainable), the proposed method for explaining the constancy of the speed of light—Chapter 5.3, “When Light Travels Faster Than Light”—allows the experiment to be conducted already today. This method is based on measuring the signal's transit time using a relativistic relay, which can increase or decrease the signal's transit time relative to the time theoretically predicted by the theory of relativity—as shown in Fig. 12—and, for example, in the case of a relay on a spacecraft traveling toward the moon, the estimated change in the signal's flight time should amount to a few microseconds, increasing or decreasing the signal's flight time depending on the direction of the relay's motion.

Regardless of the model's future, it should be emphasized that by changing the coordinate system used to describe reality, we have indeed succeeded in significantly simplifying the description of reality itself and in unifying the description of phenomena that were previously treated as separate phenomena.

Therefore, to the question posed in the title—“Does the formulation of a unified field theory require a fundamental change in the model of reality?”—we can answer with a high degree of certainty: YES.

However, in practice, the new model of reality changes almost nothing in terms of the results of calculations and observations. On the other hand, in cases where it becomes necessary to solve new problems or develop new designs and solutions, the existing description of the problems may not suffice, and a deeper understanding of the problems will be needed, along with a description of new phenomena previously unimaginable—phenomena that, as this article demonstrates, can be described by the new model.

7 Appendix – More detailed descriptions of selected issues

7.1 Mutual observation of bodies

Given the particle-wave equation (12), we can assume that the interactions are not transmitted in the form of free particles but rather are transferred in the form of disturbances propagating along

the wave ridges with a phase velocity equal to unity. The propagation of a quantum along the ridge of a wave particle is similar to the transmission of an analog radio signal, where the signal is a modulation of a carrier wave. Specifically, such a disturbance moves with the phase of the wave along the ridge of the observed wave, and then, from the point where the wave ridges meet, it moves along the ridge of the wave being the observer. In this case, regardless of the relative motion of the two bodies—that is, the angle between the wave trajectories in E4—the sum of the paths traveled by the signal along the trajectories of the sending and receiving wave bodies is always equal to the sum of the paths traveled by the signal along the ridges of these wave bodies. This is a constant value, and the quotient of these paths is always equal to unity. This situation is shown in Fig. 14.

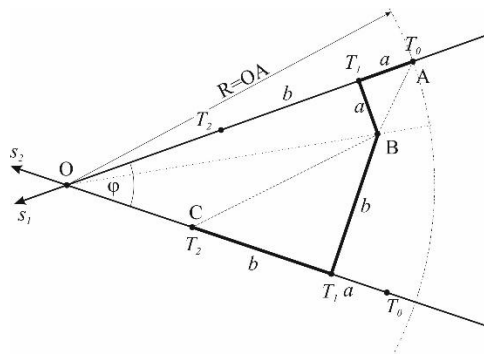


Fig. 14 Transmission of information between particle-waves. The disturbance sent at point A moves along the s_1 trajectory of the wave-particle by a distance equal to segment a and along the ridge of the signal-emitting wave-particle also by a distance a and reaches point B. At point B, it transfers to the ridge of the particle-wave receiving the signal and travels along this ridge by a distance b while also traveling a distance b along the trajectory s_2 of the particle-body receiving the signal. As a result, the signal reaches the center of the particle-wave at point C. NOTE: in this figure, the particle-waves at the time T_0 , (i.e., at the time of sending the disturbance) are at different distances from point O. Particle-wave s_1 is at a distance $R=OA$ from point O while particle-wave s_2 is at a shorter distance than particle-wave s_1 .

According to the figure, at the initial moment, both particle waves are on their trajectories at time T_0 . The particle wave moving along trajectory s_1 sends a signal, which is the disturbance, at point A and in absolute time T_0 . From this point, the particle carrying this disturbance travels along the trajectory s_1 at a distance a , and at the same time, the disturbance travels along the ridge of the particle wave—in the direction perpendicular to the trajectory—at a distance a . Now, the resulting path of the disturbance in E4 is segment AB. At point B, the ridges of the sending and receiving particle waves intersect. From this point, the disturbance moves along the ridge of the particle-wave moving along the trajectory s_2 , receiving a signal that is perpendicular to the trajectory of the particle-wave s_2 and travels along this ridge at a distance equal to b . Moreover, the particle wave s_2 travels along its trajectory at distance b . Finally, the signal reaches the particle wave s_2 at point C and at absolute time T_2 . That is, the signal now travels along the resultant trajectory BC in E4. Thus, between sending and receiving the signal, the particle waves travel a distance equal to $T_2 - T_0$ equal to $a + b$ along their trajectories, which are perceived as the axes of time; at the same time, the distance between the particle waves measured along their ridges is also equal to $a + b$. Thus, from the point of view of both particle waves, the velocity of the transfer of interactions is

$$V = \frac{a+b}{a+b} = 1 \quad (48)$$

This velocity is independent of the angle φ between the bodies' trajectories; that is, it does not depend on the bodies' relative velocity.

Thus, the velocity of transmission of interactions from the point of view of both bodies is equal to unity and does not depend on the relative velocity of the bodies.

Bodies cannot observe the trajectory of the signal; they only observe the moment when the signal is sent at point A and received at point C. Hence, the signal trajectory from the observer's point of view is a segment AC. Since the speed of light is constant and equal to unity, from the observer's point of view, the path measured along the direction interpreted by the observer as a space dimension defining the distance between particles at the moment the first particle wave sends a signal should be equal to the path the observer takes along its trajectory from the moment of sending to receiving the signal. Thus, the segment AC should be the base of the equilateral triangle ACD, whose arms are the time and space coordinates, with the time coordinate along the observer's trajectory, which the observer interprets as his time axis (this is the case for inertial motions).

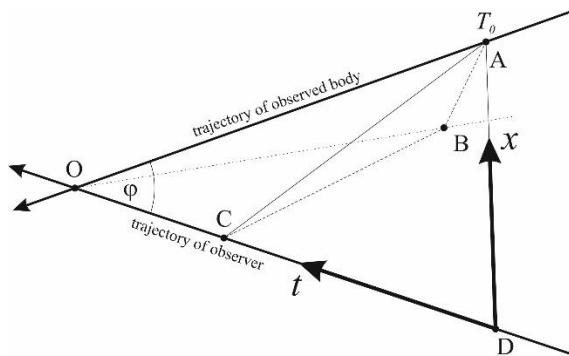


Fig. 15 Directions in E4 interpreted by the observer as space-time dimensions. The directions x, t are the arms of an equilateral triangle ACD with base AC.

This triangle is shown in Fig. 15. However, in this figure, the direction interpreted as a space coordinate is not perpendicular to the trajectory of the observed body, as shown previously in Fig. 1. As a consequence, the well-known relativistic relations that result from the space axis of the observer's coordinate system being perpendicular to the trajectory of the observed body are not fulfilled either.

Therefore, what is the problem?

7.1.1 Simultaneous events

Note that in Fig. 5, the bodies at the time of signal transmission are at the same time T_0 at different distances from point O, where the trajectories intersect. This means that body s_2 , which is closer to point O, will reach it earlier than body s_1 . Thus, the bodies at point O will not meet.

However, in Fig. 1, the intersection point of the trajectory is the point where the bodies meet.

Let us now redraw Fig. 15 according to the assumption made for Fig. 1; that is, assume that at the point where the trajectories intersect, the bodies will meet. If the bodies are to meet at point O, then they must be equidistant from point O when the signal is sent. In Fig. 16, the two bodies are at time T_0 at a distance from point O equal to OA.

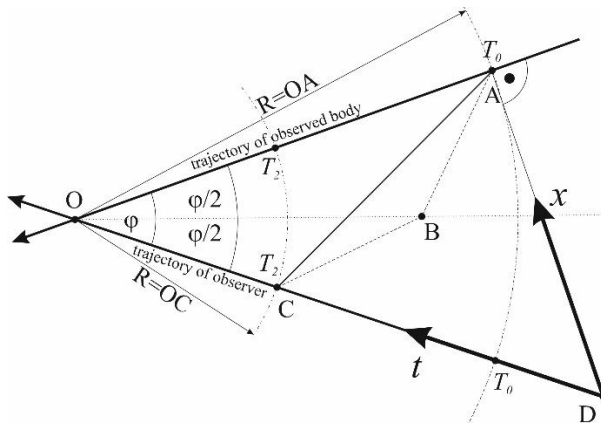


Fig. 16: At the time of sending the signal, both bodies are on their trajectories at points equidistant from point O (at a distance R). Thus, upon reaching point O, the two bodies will meet. Only in this case the direction interpreted as the space axis of the observer is perpendicular to the trajectory of the observed body. At the same time, the point B at which the signal transfers from the ridge of the body sending the signal to the ridge of the body receiving the signal lies on the bisector of the angle φ .

When the signal reaches body s_2 (at point C), i.e., at time T_2 , the bodies are also at an equal distance from point O, and upon reaching point O, both bodies have four identical space coordinates and identical absolute time values T .

In such a case, the direction in the drawing interpreted as the space axis of the observer's coordinate system is perpendicular to the trajectory of the observed body, and the relativistic relations described by formulas (1), (2), and (3) are satisfied; a proof of this can be found in the literature [59].

In such a case, bodies meet at the intersection of trajectories—that is, they have a point with five identical coordinates—four of space and one of time—described in absolute time. This corresponds to events that we call **absolutely simultaneous events**. However, what in the case when we do not have a flatness problem, as in the drawings, only has a spatial problem—with motion in three-dimensional space, where it may be impossible to determine the intersection point of the trajectory?

Note that in the case of absolutely simultaneous motions, the point at which the wave ridge exchange interactions intersect is always the bisector of the angle between the trajectories of the bodies—in Fig. 16, point B—or in other words, the point at which the wave ridges intersect is equidistant from the trajectories of these waves.

Therefore, we can now define absolutely simultaneous events—for now, only for inertial motions (along rectilinear trajectories)—without having to determine the common point of the trajectory.

In the case of inertial motions, absolutely simultaneous events occur in which, at each point in the trajectory, the point at which the wave ridges intersect is equidistant from the trajectories of these bodies, with the distances measured along the wave ridges.

Thus, for absolutely simultaneous events, as shown in Figs. 1 and 16, we have satisfied the relativistic principles known from our environment, whereas for events that are not absolutely simultaneous, such as the events in Figs. 14 and 15, the relativistic principles known to us are not satisfied.

Are absolutely simultaneous events the rule throughout the Universe or is it just an exception, and are we just dealing with absolutely simultaneous events in our environment?

In other words, absolutely simultaneous events occur when bodies meet from time to time. We, leaving for work and returning, sending rockets to the moon or Mars, live in a world of absolutely simultaneous events. Additionally, orbital motions and movements of planets around the Sun can also be presented as satisfying the condition of absolute simultaneity, although here, I will not

longer elaborate. Additionally, the Big Bang hypothesis suggests the absolute simultaneity of all events because, according to this hypothesis, the trajectories of all bodies had a common beginning. Therefore, theoretically, all events in the Universe should be absolutely simultaneous.

However, it is very likely that during the expansion of the Universe and the various interactions and motions associated with it, some regions of the Universe ceased to meet the condition of absolute simultaneity with respect to bodies in our surroundings, and in that case, astronomical observations of distant areas of the cosmos should yield different results than those predicted by the Theory of Relativity.

At this point, the following question can be posed: are there any forces or principles of behavior that result in the preservation of absolute simultaneity, and can this absolute simultaneity be disrupted? Thus far, I do not know the answer to this, but in the case of space travel, especially at relativistic speeds, this could be a key problem to ensure that we reach our destination.

To summarize:

As shown in Fig. 1, we have two types of simultaneity: a simultaneous event from the observer's point of view lying in a direction in E4 interpreted as the space axis of the observer, where the notion of simultaneity depends on the choice of the observed body, and absolute simultaneity, which does not depend on the choice of the observer and whose conservation is a condition for the relativistic relations valid in our environment.

7.2 Mechanism to change the time

Thus far, when considering inertial motions, we have dealt with only the rectilinear trajectories in E4. Having defined the concept of absolutely simultaneous events, for inertial motions, we can consider noninertial motions.

Noninertial motions, as seen from observation and everyday life, also preserve the absolute simultaneity of events.

To discuss this issue in more detail, let us first consider the example of the paradox of twins, in which one of the twins goes on a rocket trip and returns younger than the twin who stayed on Earth.

We consider the problems of instantaneous takeoff and landing of the rocket and an instantaneous change in the direction of the rocket. Here, we focus only on constant-speed flight and turning the rocket back.

The trajectories of the rocket and Earth at E4 are shown in Fig. 17.

In Fig. 17, the rocket is shown not as a point mass body but as a wave. The change in the rocket velocity shown in E4 is equivalent to a change in the direction of the wave representing the rocket in E4. In the case shown in Fig. 17, the change in the direction of the rocket's motion corresponds to the rotation of the wave representing the rocket by an angle φ_{BC} corresponding to the double speed of the rocket (assuming that the speed of the rocket V in both directions is the same).

$$\sin(\varphi_{BC}) = 2V \quad (49)$$

where the wave rotates relative to point O on the time axis of the Earth-bound coordinate system (according to the principles shown in Fig. 1).

Owing to this, we can see that the immediate change in the direction of the rocket's motion causes the rocket, represented as a wave, to remain in the same state at both point B and point C; i.e., it

simply does not change phase during its transition from point B to point C. Thus, the BC arc is the section of the rocket's trajectory where the phase of the wave in E4 describing the rocket does not change. Since in E4, the measure of time is the number of vibrations performed by the wave between two points, the phase constancy on the BC segment means that the wave does not perform any vibrations on this segment; i.e., the time indicated on the rocket's clock, equal to the number of wave vibrations, is equal to zero on the BC segment.

Of course, if we treat the arc BC as the world line (of a mass body), then assuming zero time flow on a nonzero segment of the world line is nonsense. However, here, we describe the motion of the rocket via a trajectory in which the proper time is not a property of reality but is measured via the number of wave cycles; thus, theoretically, on a nonzero segment of the trajectory, the proper time of the body may be equal to zero. In other words, all the points on the arc BC are in the same state. The possibility of points distant from each other in space being in an identical state was once criticized by Einstein, Podolsky, and Rosen [65] in their critique of quantum mechanics. Today, it is an accepted fact known in quantum mechanics as quantum entanglement.

Incidentally, the instantaneous speed change is not a physical phenomenon and is mentioned here only to illustrate the basics of the time change mechanism discussed later. However, regardless of whether we describe the problem according to the E4 model or in Minkowski spacetime, an instantaneous change in velocity is associated with infinite acceleration and zero-time flow during the turn, whereas when this problem is presented in Minkowski spacetime, points B and C coincide—Fig. 17b.

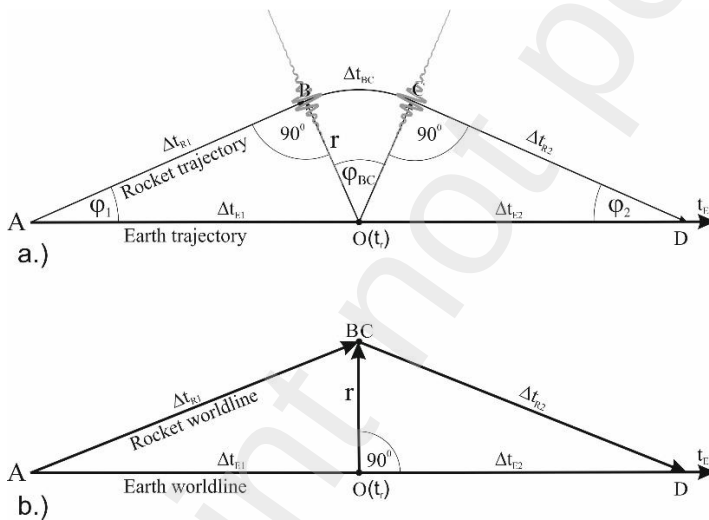


Fig. 17 The twin paradox in E4 – Fig. a and in Minkowski spacetime – Fig. b. The rocket turns back immediately. At the moment of turning back, the rocket in E4 travels along the arc BC, on which the wave phase does not change, which corresponds to zero time flow in the rocket. In Minkowski spacetime – fig. b, points B and C coincide, which also corresponds to zero time flow. Figure a. shows that in order to describe phenomena in E4 space, it is necessary to accept motion along arcs along which the wave phase does not change.

According to the previous interpretation that states that, in the case of rectilinear trajectories, the length of the trajectory is a measure of the time indicated by the clock of an observer moving along the trajectory, we see that the lengths of segments AB and CD of the rocket's trajectory and segment AD of the Earth's trajectory are measures of the indications of the clocks in the rocket and on Earth. However, as already stated, we assume that the time of flow in the rocket on arc BC (Fig. 17a) is equal to zero.

$$\Delta t_{BC} = 0 \quad (50)$$

Thus, the flow of time on the Earth's surface is equal to:

$$\Delta t_E = \Delta t_{E1} + \Delta t_{E2} \quad (51)$$

The time flow in the rocket is equal to:

$$\Delta t_R = \Delta t_{R1} + \Delta t_{BC} + \Delta t_{R2} = \Delta t_{R1} + \Delta t_{R2} \quad (52)$$

On the basis of the dependence between the sides of the triangles ABO and OCD (as shown before in Fig. 1):

$$\Delta t_{Ri} = \Delta t_{Ei} \cos \varphi_i = \Delta t_{Ei} \sqrt{1 - V_i^2} \quad (i = 1, 2) \quad (53)$$

Formula (31) can be written in the following form:

$$\Delta t_R = \Delta t_{E1} \sqrt{1 - V_1^2} + \Delta t_{E2} \sqrt{1 - V_2^2} \quad (54)$$

This, incidentally, is consistent with the results of the experiments.

Thus, to obtain relation (54) from equation (52), which expresses the total flow of time in the rocket system, one must assume that the flow of time in the rocket on the BC trajectory section equals zero. This is the price we have to pay for using the Euclidean model of reality. In the case of the Minkowski spacetime model, points B and C in Fig. 17 coincide, so there is no problematic arc BC; however, we must use a pseudo-Euclidean metric that forces the deformation of the time axis of systems in motion.

This assumption does not agree with the previous definition of absolute simultaneity for inertial motion; however, in that case, we must make an additional assumption—this time, only for noninertial motions:

We assume a change in the direction of motion, as shown in Fig. 17; that is, the motion of the body along the arc of a circle with the center on the trajectory of the observer's inertial system is as follows:

1. **is instantaneous in the system of an inertial observer,**
2. **does not cause the flow of time in the system of the body observed, along the part of the trajectory that travels along this arc,**
3. **preserves the principle of absolute simultaneity of motion.**

One may be doubtful about the validity of such a bottleneck assumption, but this assumption is necessary for ensuring the consistency of the model and its agreement with measurements.

On the other hand, as I wrote previously, the arc BC in Fig. 17 represents points with a constant phase of a given wave, i.e., points distant in space whose state is identical. This phenomenon is very similar to quantum entanglement. Note that if two particles are created in E4 at the same point and time, even if they may move at different speeds in the laboratory system, in E4, they always remain on a sphere with the same radius for both particles and the same phase—analogue to points B and C in Fig. 17. Therefore, regardless of distance, such particles should always behave as if they were at the same point. This may be too far from reaching a conclusion and too speculative at this stage, but perhaps time dilation and quantum entanglement, although they appear to us as completely different phenomena, may have the same source.

To conclude, in Figure 17, I would like to add that, at point O in Figure 17, we address a change in the direction of wave motion in E4. This is an unrecognized and puzzling issue, especially since the ridge of our wave extends to infinity, and the problem of changing the direction of motion of the wave, i.e., its rotation in E4, is not yet known, and we must rely for the moment only on the analysis of mathematical relationships.

However, this seemingly absurd assumption about the possibility of instantaneous motion of a body along the arc of a circle with a center at a point on the observer's trajectory provides concrete benefits because it immediately resolves issues that have hitherto raised many doubts, namely,

1. During uniform motion, observers register a mutual dilation of time in the neighbor system, and this effect is mutually symmetrical.
2. The actual dilation time occurs in the system of observers who change the value or direction of the velocity.
3. Although the determining factor for the actual occurrence of time dilation is the change in velocity, the observed time dilation is solely a function of the velocity of the body whose system time has slowed down and is fully consistent with the predictions of the special theory of relativity according to formulas (2) and (3).

Thus, according to the above reasoning, **it is not true that time slows down in systems in motion**, as has been assumed since the turn of the 19th century. **Time slows down as a result of a change in the velocity of bodies, and only the value of the change in time depends on the difference in the velocity of bodies.** Hence, the results of experiments in which the change in time was only a function of the speed of the body relative to the observer were misinterpreted as evidence of the actual change in the flow of time of bodies only as a function of uniform motion, which, in light of the approach proposed here, turns out to be untrue.

This is a makeshift mechanism for ensuring the consistency of the model. However, a thorough explanation of the mechanisms occurring during the change in the direction of body-wave motion at E4 is crucial for understanding the mechanism responsible for the time delay.

Having solved the problem of the absolute simultaneity of events both for sections with a constant velocity of bodies and for points where there is a change in the direction of body-wave motion, we can describe noninertial motion—only in the system of an inertial observer.

To describe noninertial motion, we break the trajectory of noninertial motion in E4 into a series of short rectilinear segments dt' analogous to segments AB and BC in Fig. 17. We connect these segments with arcs dl corresponding to the arc BC in Fig. 17. Analogous to Fig. 17, the flow of time in the body system in motion along such arcs is zero.

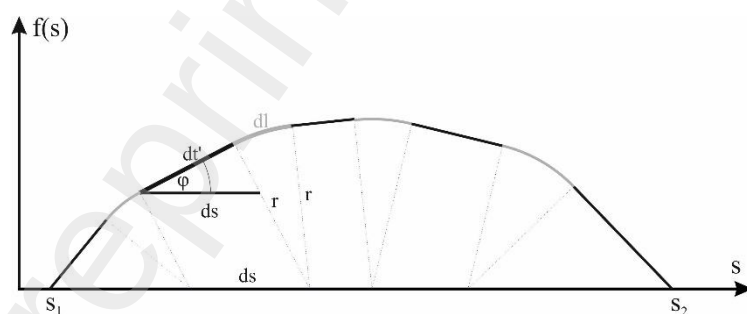


Fig. 18 The trajectory of a noninertial body described in E4 by the function $f(s)$ represented as a complex of linear segments, and arcs of circles (with centers on the trajectory of the inertial observer) along which the flow of time is zero.

Thus, the flow of time in a system in motion is equal:

For dt' segments according to the rules adopted for the rectilinear trajectories in Fig. 1 and with formulas (2) and (3),

$$dt' = ds \cos\varphi \quad (55)$$

For time segments dl

$$t' dl = 0 \quad (56)$$

Now, if time $t = s_2 - s_1$, flow in the observer system in Fig. 18; then, for a curvilinear trajectory, as in Fig. 18, consisting of pairs of segments dt' and dl , we can determine the flow of time with the formula

$$t' = \int_{s_1}^{s_2} dt' + \int_{s_1}^{s_2} t' dl = \int_{s_1}^{s_2} \cos\varphi ds = \int_{s_1}^{s_2} \sqrt{1 - V^2} ds = \int_{s_1}^{s_2} \frac{ds}{\sqrt{1 + \left(\frac{\partial f(s)}{\partial s}\right)^2}} \quad (57)$$

Moreover, the length of the trajectory described by the function $f(s)$ in Fig. 18 measured at E4 is

$$s_f = \int_{s_1}^{s_2} \frac{ds}{\cos\varphi} = \int_{s_1}^{s_2} \sqrt{1 + \left(\frac{\partial f(s)}{\partial s}\right)^2} ds \quad (58)$$

The length of the trajectory describing the track of the noninertial particle in E4 (58) is greater than the corresponding length of the trajectory of the inertial observer, which is equal to

$$t = \int_{s_1}^{s_2} ds = s_2 - s_1 \quad (59)$$

A shorter time then passes in the system of a body on a curvilinear trajectory (57) than in the system of an inertial observer (59). In the theory of relativity, this effect is explained by the deformation of space-time according to the spacetime interval equation (more about the sense of the space-time interval according to the E4 model in [63,65]). Moreover, in E4, temporal dilation is not related to space deformation because space is constantly undisturbed here (local disturbances describing waves are ignored), and waves in the same region from bodies moving differently will have their period of oscillation unchanged or otherwise altered, depending not on the metric of space itself but on the motion of these particular waves and interactions with other waves. The slowing down of time on a curvilinear trajectory is related to the mechanism assumed in this work for changing the direction of motion of a wave, the consequence of which is that the period of the wave increases when its direction of motion changes. In the theoretical case shown in Fig. 17, where the change is instantaneous, the wave does not change phase on the BC curve; thus, the period of the wave is infinite in this case. Of course, this is a theoretical, not physical, example intended only to present the phenomenon.

In the case shown in Fig. 18, the trajectory is a curved line, and different sections of the trajectory (of length $dt' + dl$) inclined at different angles to the observer's trajectory can be assigned different values (55). Here, the length of the trajectory is no longer directly a measure of the particle's proper time. Thus, for noninertial motions, the time in the particle system is measured not by the length of the trajectory but by the number of periods of the particle. The curvature of the trajectory and the resulting change in the flow of time are not related here to the deformation of space but rather to the change in the number of body-wave oscillations. Thus, the problem of noninertial motion from Fig. 18 should be described via formula (13), which is prescribed for time and is measured as the number of oscillations of the particle. In this case, we describe time by the number of oscillations of the particle along a given path; that is, we insert the path expressed by the number of oscillations of the particle into formulas (57) and (59):

$$s = \frac{2\pi}{\omega} n \quad (60)$$

In this case, equation (38), which describes the time in the observer's system, takes the form

$$t = \frac{2\pi}{\omega} \int_{n_1}^{n_2} dn = \frac{2\pi}{\omega} (n_2 - n_1) \quad (61)$$

For a body moving along a trajectory $f(s)$ relative to an inertial observer, the elapsed time is equal to

$$t' = \frac{2\pi}{\omega} \int_{n_1}^{n_2} \frac{dn}{\sqrt{1 + \left(\frac{\omega}{2\pi} \frac{\partial f(\frac{2\pi}{\omega} n)}{\partial n} \right)^2}} \quad (62)$$

Thus, although the trajectory of a body moving in noninertial motion (58) is longer in E4 than the trajectory of an inertial body (59), the noninertial body performs fewer oscillations along this path than does the inertial body along its entire path. What is responsible for the smaller number of oscillations is not the deformation of space, as assumed by the theory of relativity, but the mechanism of changing the direction of wave motion in E4, which, according to the mechanism shown in Fig. 17 and in formulas (50)-(54), should reduce the number of oscillations along the curvilinear trajectory of the particle.

To date, the scheme shown in Fig. 18 has been developed only for $f(s)$ described by 2nd-degree polynomials with a maximum. The function $f(s)$ works in E4 coordinates. In addition, in Fig. 18, we assume a trajectory, which allows the trajectory to be represented as a series of segments and arcs of a circle with the center located on the observer's trajectory. On the other hand, at some point, if the radius of curvature of the trajectory is smaller than the radius of such an arc, then the body moving along such a curved trajectory and the inertial observer will be at point s_2 at different times T ; that is, the bodies will not be able to meet longer. This is an unsolved problem that is likely to arise in the planning of space travel when it is necessary to plan an appropriate trajectory for the rocket, ensuring that it reaches its destination not only at the desired location but also at the desired time. However, with the accelerations achieved today, all the trajectories of our rockets are nearly straight lines, and this kind of problem is unlikely to be of concern.

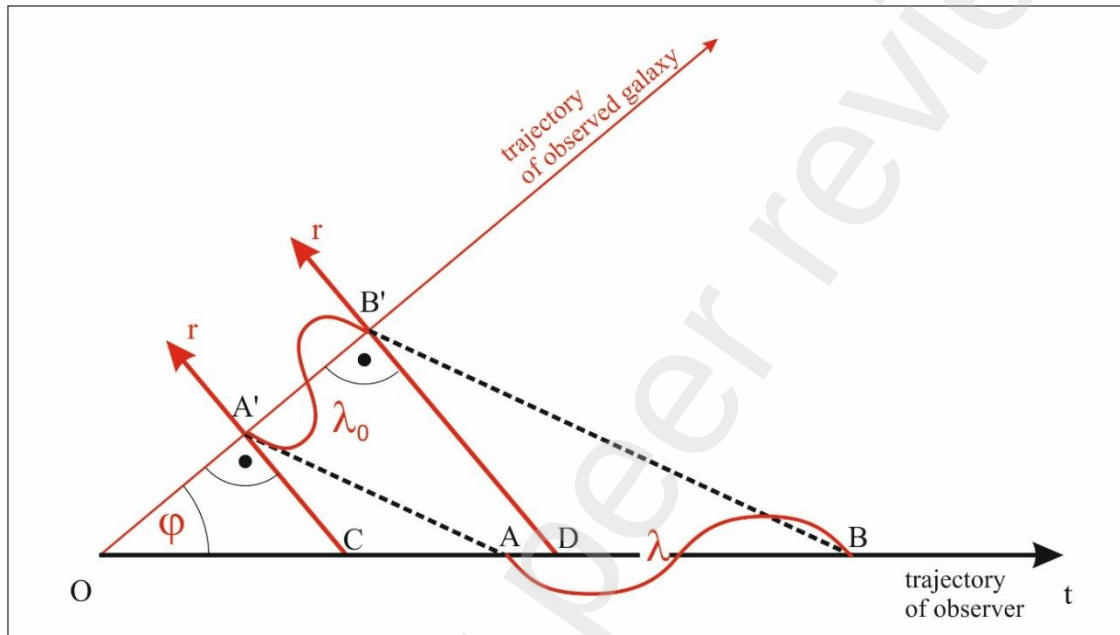
The mechanism outlined above explains the problem of time dilation in an unambiguous and consistent way, resolving any doubts arising in earlier discussions [46] but at the expense of introducing a mechanism for changing the frequency of a wave when its direction of motion changes in E4. This mechanism has not yet been validated and requires additional explanation; however, it must be adopted to ensure the consistency of the model.

Note that in the case of noninertial motions, in principle, the directions of the time and space dimensions cannot be determined as they were in the case of inertial motions. While the selected directions in the E4 space in the case of inertial motions for a specific pair of bodies, i.e., the observer-observed body, were possibly associated with the observed dimensions of time and space (Fig. 1), in noninertial motions, in principle, we discuss trajectories in E4. In such cases, at each point of the trajectory, the directions interpreted as time dimensions and space dimensions are different; thus, for noninertial motions, we cannot discuss temporal or spatial dimensions. Thus, any noninertial problem should be considered only in E4 coordinates, and only later can the data obtained, such as the data in Fig. 18, be plotted on a graph showing the values of time and distance we measure in our coordinate system where we assume the perpendicularity of space-time dimensions; however, this is only a mathematical picture that does not give us an idea of the actual shape of reality.

Thus, the dimensions of time and space that we observe are not properties of physical reality but only directions in E4 that we imagine as dimensions of time and space on the basis of observations of our nonrelativistic environment.

7.3 Derivation of the formula for the redshift of the radiation of galaxies according to the proposed model of reality.

The problem of the galaxy and the observer is shown in Figure 19. The figure shows the time axes of the observer - t and the observed galaxy t' . The space axes of the observer's system are plotted perpendicular to the trajectory of the observed body, that is, in our case, the galaxy. The galaxy sends out a wave of length λ_0 , which in the figure is represented by segment $A'B'$. The



wave travels at the speed of light (according to Figure 3); that is, the beginning and end of the wave travel along different paths— $A'A$ and $B'B$, respectively. The wave that reaches the observer on Earth is longer than the sent wave; let us denote it as λ , and in Figure 19, it is represented by segment AB .

Fig. 19 Time axes of the galaxy and the observer on Earth. The spatial axes of the observer are plotted perpendicular to the time axis of the observed galaxy. $A'B'$ denotes the segment equal to the wavelength sent by the galaxy, and segment AB denotes the wavelength received by the observer on Earth. The segments $A'A$ and $B'B$ denote the paths of the beginning and end of the wave traveling at the speed of light, respectively.

Figure 19 shows that the condition for the constancy of the speed of light is as follows:

$$c = \frac{\Delta r}{\Delta t} = \frac{A'C}{CA} = \frac{B'D}{DB} = 1 \quad (63)$$

requires the following relationships to be true between the sections representing the elapsed times in the Earth and galactic coordinate systems and the distances measured from the galaxy:

$$OA = OC + CA \quad (64)$$

$$CA = CA' \quad (65)$$

In turn, the following relationships follow from triangle $OA'C$:

$$OC = \frac{OA'}{\cos\varphi} \quad (66)$$

$$CA' = OA' \tan\varphi \quad (67)$$

$$OA = OC + CA' = \frac{OA'}{\cos\varphi} + OA' \tan\varphi = OA' \frac{1+\sin\varphi}{\cos\varphi} \quad (68)$$

In turn, it follows from the definition of velocity (1) that:

$$OA = OA' \frac{1+V}{\sqrt{1-V^2}} = OA' \frac{\sqrt{1+V}}{\sqrt{1-V}} \quad (69)$$

Similar relationships hold for the triangle $OB'D$:

$$OB = OB' \frac{\sqrt{1+V}}{\sqrt{1-V}} \quad (70)$$

Since the segments AB and $A'B$ represent the wavelengths of the received wave λ and the transmitted wave λ_0 , respectively, we see the following:

$$\lambda = OB - OA = (OB' - OA') \frac{\sqrt{1+V}}{\sqrt{1-V}} = \lambda_0 \frac{\sqrt{1+V}}{\sqrt{1-V}} \quad (71)$$

Therefore, we obtain the following:

$$\frac{\lambda}{\lambda_0} = \frac{\sqrt{1+V}}{\sqrt{1-V}} \quad (72)$$

When describing the redshift, we use the z symbol defined as follows:

$$z = \frac{\lambda}{\lambda_0} - 1 \quad (73)$$

We eventually obtain the relativistic formula for redshift, which is also a description of the relativistic Doppler effect.

$$1 + z = \frac{\sqrt{1+V}}{\sqrt{1-V}} \quad (74)$$

The derivation of the redshift formula does not require any knowledge of the theory of relativity and results in a simple triangle solution.

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In the E4 model and the example provided, what if "t1" observes "t2," and once the information arrives, it sends it to the observer "t"? The angle to "t1" is smaller, so this 'relayed' information will arrive faster than the original from "t2." Thank you in advance for your response.
Original text: W modelu E4 i zaprezentowanym przykładzie, co jeśli "t1" będzie obserwować "t2" i jak dotrze informacja to wyśle informację do obserwatora "t"? Kąt do "t1" jest mniejszy więc ta "przekazana" informacja dotrze szybciej niż oryginał od "t2". Z góry dziękuję za odpowiedź"